Postulates, Parent Geometry, and Dynamical Sectors of Complex Relativity

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Abstract

Complex Relativity begins with a common light cone for real clocks and a complex metric, and with real measured free momenta represented by one compatible mass shell. This article organizes four parent-theory studies in the order of their logical dependencies. The postulates first reduce the complex metric on a real slice to a single complex conformal factor. A complex structure and a real tangent projector then define a maximally real distribution, an auxiliary Born structure, and a Lorentzian clock operator that reconstructs the physical one-time bulk metric. On this geometry, a specified Einstein–response model admits a common scalar parent with a round S^5 target. At fixed couplings, its poles recover the full independent bulk theory B and its invariant equator recovers the full defect theory D. Their exact four-dimensional reductions possess the local constrained mirror-holomorphic representation H under explicit analyticity assumptions. The defect has finite gravitational normalization and a normalizable tensor zero mode, but a gapless continuum and no free massive-particle confinement. Its added odd scalar has a sharp nonnegative threshold, while its full coupled stability remains unresolved. We retain the regular local extension of the flat defect and the finite-affine-distance curvature obstruction for a rolling-scalar uplift. Further results classify all stationary target loci, invariant subspheres, Einstein products, offset and tilted real slices, and a twisted normal distribution. An explicit joint J/P trial action and exact nonuniqueness counterexamples delimit the stronger claim of a uniquely determined geometric mother theory. The resulting framework is an exact classical construction with stated branches and proofs; it is not presented as a completed quantum unification.

Scope: The physical status of each construction is stated at the point of use: algebraic geometry, an exact sector of the scalar parent S_*, a conditional analytic representation, or an unclosed geometric trial system. No low-energy expansion defines the bulk reductions.

CR manuscripts: github.com/Gauss-K/Complex-Relativity

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How to read this synthesis

This article integrates the audited B/D/H manuscript, the common S^5 parent, the geometric-variation analysis, and the parent-geometry study [1, 2, 3, 4]. The order is deliberate: observations determine admissibility conditions; geometry identifies the fields and their relations; an action specifies dynamics; exact sectors then test that action. A conclusion at one stage is not silently promoted to the next.

The old S^4 defect action has no constant zero-energy stationary point and therefore cannot itself recover B by freezing its defect field. The later S^5 construction resolves that obstruction by enlarging the field space. Both statements are retained, with their different domains made explicit. The common scalar action does not contain kinetic equations for J or P. The larger action that does vary them is a trial system, and its simplest positive-fiber kinetic terms fail on the unmodified old defect. The distinction is essential to the integrated account.

The body follows six stages: P1/P2; parent geometry; parent dynamics; B, D, and H in that order; other target sectors and spacetime slices; and the resulting research problem. The appendices retain the analytic variational argument, canonical constraints, inverse construction, radial-lapse check, and reproducibility record. Common definitions occur once; later sections use the same action, metric, and multiplier conventions.

Symbol	Meaning	Distinction
g, γ	Real clock metric and induced complex metric	$\gamma = (1 + \mathrm{i}\chi)g$ on the slice
$\widehat{g}$	Physical eight-dimensional metric	Signature $(1, 7)$
J, P	Complex structure and rank-four real projector	Their algebra does not fix their dynamics
$q, \eta_{\mathrm{B}}, \omega$	Auxiliary Born forms	Positive, neutral, and skew respectively
A	Lorentzian clock operator on $E = \operatorname{im} P$	Distinct from the warp exponent $A = \log \Omega$
X^α, N^a	S^5 parent field and S^4 equatorial field	Targets are not spacetime dimensions
$Q, \mathfrak{h}, C_j$	Dimensionless CR fields, their target, response parameters	Every response parameter is uppercase $C_j > 1$
Λ, λ_D	Parent and defect unit-constraint multipliers	$\lambda_D = \Lambda + 24/\ell^2$ on D
$H, \Gamma_\pm$	Holomorphic clock extension and reflected metrics	H also names the corresponding construction
$\mathcal{M}, \Lambda_H^{ab}$	Readout and mirror tensor multipliers	Different equations and on-shell behavior

Conventions: Units set the speed of light and $\hbar$ to one. Slice signature is $(-+++)$; the bulk has one negative and seven positive directions. Greek indices run over four real-slice coordinates. Capital Latin spacetime indices run over all eight coordinates, while $A, B = 1, \ldots, 4$ label normal coordinates and the first four order-parameter components. The parent target index is $\alpha = 1, \ldots, 6$, and $a = 1, \ldots, 5$ denotes its equator. Later local geometric examples use paired normal labels y^μ, $\mu = 0, \ldots, 3$; this is the relabeling $y^\mu = y^{A=\mu+1}$, not a second time coordinate. The scalar count is $n = 1 + 2n_{\mathrm{p}}$. The radius r, the potential parameter η, the Minkowski tensor $\eta_{\mu\nu}$, and the Born form η_{B} have separate roles.

Part I

P1/P2 and real observations

1 The postulates and their exact algebraic consequences

1.1 The two postulates on the real slice

Let M be an oriented, time-oriented four-manifold with a real Lorentzian metric g of signature $(-+++)$. Write the induced complex symmetric metric as

$$\gamma_{\mu\nu} = g_{\mu\nu} + \mathrm{i}b_{\mu\nu}, \qquad b_{\mu\nu} \in \mathbb{R}. \tag{1.1}$$

Here g measures the proper times of real clocks. We use the following operational form of the CR postulates.

P1: common light cone. Every real tangent that is null for g is null for γ. Since the latter condition is complex, its imaginary part requires $b(v, v) = 0$ whenever $g(v, v) = 0$.

P2: real free-particle momentum. The measured free covector p is real on M, its complex extension is the natural one supplied by the real slice, and its complex quadratic mass-shell representation must encode the same real free-particle constraint.

Lemma 1.1 (Algebraic light-cone reduction)**.** *For real symmetric b in four dimensions, P1 implies*

$$b = \chi g, \qquad \gamma = Fg, \qquad F = 1 + \mathrm{i}\chi, \quad \chi \in \mathbb{R}. \tag{1.2}$$

Proof. In a g-orthonormal frame, use null vectors $(1, \mathbf{n})$, where $|\mathbf{n}| = 1$. The equation $b_{00} + 2b_{0i}n_i + b_{ij}n_in_j = 0$, compared at $\mathbf{n}$ and $-\mathbf{n}$, gives $b_{0i} = 0$. The quadratic form $b_{ij}n_in_j$ is constant on the unit sphere, hence $b_{ij} = -b_{00}\delta_{ij}$. Set $\chi = -b_{00}$. □

Thus ten real components of b reduce to one scalar for fixed g: nine independent algebraic restrictions. On this branch P2 has the compatible representative

$$\mathcal{Q}_{\mathrm{R}} = g^{\mu\nu}p_\mu p_\nu + m^2, \qquad \mathcal{Q}_{\mathbb{C}} = F^{-1}\mathcal{Q}_{\mathrm{R}}. \tag{1.3}$$

For finite real χ, $F \neq 0$; there is one real mass shell, not two independent real equations. We exclude the singular massless vertex $p = 0$ when discussing a regular constraint surface. The real worldline action is

$$S_{\mathrm{probe}} = \int \left(p_\mu \dot{x}^\mu - \frac{e}{2}\mathcal{Q}_{\mathrm{R}} \right) \mathrm{d}\tau. \tag{1.4}$$

Its free trajectories are g-geodesics. No independent requirement of affine γ-geodesic motion is imposed.

The equivalence in (1.3) is an equivalence of zero sets on the real phase space. Writing $q = \mathcal{Q}_{\mathrm{R}}$,

$$\mathrm{Re}\,\mathcal{Q}_{\mathbb{C}} = \frac{q}{1+\chi^2}, \qquad \mathrm{Im}\,\mathcal{Q}_{\mathbb{C}} = -\frac{\chi q}{1+\chi^2}. \tag{1.5}$$

On the regular shell, their differentials are proportional to $\mathrm{d}q$, and hence have rank one. Away from it, treating q and χ as independent local variables gives

$$\det \frac{\partial(\mathrm{Re}\,\mathcal{Q}_{\mathbb{C}}, \mathrm{Im}\,\mathcal{Q}_{\mathbb{C}})}{\partial(q, \chi)} = -\frac{q}{(1+\chi^2)^2}. \tag{1.6}$$

The two functions can therefore have rank two off shell. This causes no extra mass-shell constraint. It does rule out an unrestricted off-shell rank-one assertion. Also, $\mathcal{Q}_{\mathbb{C}}$ includes the same factor on both the kinetic and mass terms; it is not obtained by changing the inverse metric while holding an unscaled mass term fixed.

1.2 What the postulates determine

P1 fixes the complex metric's pointwise form once the real clock metric is given. P2 requires the complex representative to encode the measured real mass shell. The real worldline action above supplies the evolution prescription; equality of mass-shell zero sets alone does not fix an off-shell Hamiltonian or a complex evolution parameter.

This distinction can be expressed without introducing a second constraint. Let $a(x,p) > 0$ be a smooth real function near the regular shell, and let X_f denote the canonical Hamiltonian vector field of f. Then

$$X_{a\mathcal{Q}_{\mathrm{R}}} = aX_{\mathcal{Q}_{\mathrm{R}}} + \mathcal{Q}_{\mathrm{R}}X_a, \qquad X_{a\mathcal{Q}_{\mathrm{R}}}\big|_{\mathcal{Q}_{\mathrm{R}}=0} = aX_{\mathcal{Q}_{\mathrm{R}}}\big|_{\mathcal{Q}_{\mathrm{R}}=0}. \tag{1.7}$$

Such representatives have the same oriented, unparametrized on-shell trajectories, with the lapse absorbing the real rescaling. A genuinely complex rescaling need not generate a vector field tangent to the real phase space. This is why the real Hamiltonian is specified separately even though $\mathcal{Q}_{\mathbb{C}}$ is a convenient complex shell representative.

Neither postulate specifies a scalar potential, an independent eight-dimensional kinetic action, a complex-structure equation, or a dynamical rule selecting a real submanifold. Those questions are addressed in the next two parts as additional constructions, with explicit checks of their compatibility with the postulates.

The distinction also prevents a false double count. The imaginary conformal coefficient χ is one real field, not nine surviving traceless imaginary metric components. Conversely, a compatible complex shell does not remove the genuinely propagating normal data of an independently postulated bulk. The geometry must say which objects are analytic continuations and which are independent fields.

Part II

Parent geometry

2 The parent has a base, tangent structures, and a scalar target

The term "parent geometry" can refer to several different objects. The distinction matters here: a submanifold of a scalar target is not an extra spacetime dimension, and a four-dimensional real slice is not a truncation of target coordinates.

The geometric configuration considered here has an eight-dimensional real base $\widehat{M}$, a Lorentzian metric $\widehat{g}$ with signature $(-+++++++)$, dimensionless response fields Q^I, and an order parameter $X : \widehat{M} \to S^5$. The geometric extension [3] adds a real complex structure J and a rank-four tangent projector P. Locally the proposed geometric data are

$$(\widehat{M}; \widehat{g}, J, P; X, Q), \qquad X : \widehat{M} \longrightarrow S^5, \qquad Q : \widehat{M} \longrightarrow \mathcal{T}_{\mathrm{CR}}. \tag{2.1}$$

The maps do not turn the scalar target into additional coordinates of physical spacetime.

Object	Geometric role	What must still be specified
$\widehat{M}, \widehat{g}$	Independent one-time bulk	Field equations and boundary data
J	Complex directions, complex type	Integrability and its dynamics
P	Real tangent/normal splitting	Alignment, integrability, global existence
$X \in S^5$	Defect order parameter and sector selection	Potential, regularity of the core
$Q \in \mathcal{T}_{\mathrm{CR}}$	Clock phase and response fields	Target metric and couplings
Γ	Complex readout of real clock geometry	Holomorphicity is an extra condition

The established labels also refer to different kinds of construction. B is an independent bulk theory recovered at constant target poles. D is a defect theory recovered on an invariant target equator, with a known warped solution. H is a constrained local holomorphic representation of specified analytic four-dimensional dynamics [1]. H is not a third vacuum of the S^5 potential. There is consequently no reason to expect B, D, and H to exhaust either the target sectors or the possible real slices.

3 Geometric variables for a moving real slice

3.1 A complex structure and a tangent projector

Take an eight-dimensional real manifold with a one-time metric $\widehat{g}$. Introduce independent real endomorphisms J and P with

$$J^2 = -\mathbf{1}, \tag{3.1}$$

$$P^2 = P, \qquad \operatorname{tr} P = 4, \qquad \widehat{g}(Pu, v) = \widehat{g}(u, Pv), \tag{3.2}$$

$$PJ + JP = J. \tag{3.3}$$

Write $P_\perp = \mathbf{1} - P$; the complementary projector is distinguished from the CR scalar vector Q^I. Then $E = \operatorname{im} P$ and $F = \operatorname{im} P_\perp$ are orthogonal rank-four distributions, with $JE = F$ and $JF = E$. Work on the open branch where $\widehat{g}|_E$ is Lorentzian and $\widehat{g}|_F$ is positive definite. Specifying this branch is part of the configuration space, not a derived stability theorem.

Use the first four order-parameter components to impose alignment,

$$P^M{}_N \partial_M X^A = 0, \qquad A = 1, \dots, 4. \tag{3.4}$$

Where the four covectors $\mathrm{d}X^A$ are independent, $E = \ker \mathrm{d}X^1 \cap \cdots \cap \ker \mathrm{d}X^4$. Its leaves are level sets of these fields. The selected regular core can be

$$M_X = \{X^1 = \cdots = X^4 = 0,\ X^5 > 0\}. \tag{3.5}$$

Its tangent space is E, and $E \cap JE = 0$; it is maximally real. The embedding is determined by dynamical scalar fields rather than fixed coordinate equations.

On a constant-pole sector, all $\mathrm{d}X^A$ vanish and (3.4) no longer determines P. Merely passing through a pole value at one spacetime point does not require these derivatives to vanish. Keeping P independent avoids division by a rank-deficient gradient matrix. It does not resolve the change in constraint rank between the constant-pole and regular defect sectors; that must enter the Hamiltonian analysis. The existence of a regular core is not enforced in every solution, and global existence of the required bundle splitting is an additional question.

3.2 Integrability is a separate condition

An almost complex structure satisfying (3.1) is not automatically integrable. In the smooth category impose the Nijenhuis constraint

$$\begin{aligned}(\mathcal{N}_J)^K{}_{MN} = {} & J^R{}_M \partial_R J^K{}_N - J^R{}_N \partial_R J^K{}_M \\ & - J^K{}_R (\partial_M J^R{}_N - \partial_N J^R{}_M) = 0.\end{aligned} \tag{3.6}$$

It is tensorial and may equivalently be written with any torsion-free connection. The Newlander–Nirenberg theorem relates this condition to local complex coordinates under its regularity hypotheses [16]. We use smooth fields here, not a low-regularity extension.

For a real-analytic regular core and compatible real-analytic integrable J, an adapted holomorphic chart straightens the core to a real coordinate locus. Conjugation defines a local antiholomorphic involution fixing it. Its germ is unique once J and the core are fixed, by holomorphic uniqueness on a maximally real slice. This determines a local real structure from those inputs; it does not uniquely determine the inputs or guarantee a global involution.

4 The tangent algebra already contains a Born structure

4.1 From the real projector to three endomorphisms

Use the algebraic constraints of Section 3, with $E = \operatorname{im} P$, $F = \operatorname{im} P_\perp$, Lorentzian E, and positive F. These distributions are orthogonal and are exchanged by J. Define

$$K = 2P - \mathbf{1}, \qquad L = JK. \tag{4.1}$$

Then

$$J^2 = -\mathbf{1}, \qquad K^2 = L^2 = \mathbf{1}, \qquad JK = -KJ = L, \qquad LK = J, \quad LJ = K. \tag{4.2}$$

All three endomorphisms anticommute pairwise. This is the split-quaternion algebra. Algebraic compatibility alone gives an *almost* complex-product structure; ordinary integrability conditions

remain separate. The terminology and the integrable complex-product setting are discussed in [19].

The real eigenbundles of K are exactly E and F. Thus the physical tangent/normal distinction is encoded in an endomorphism, without selecting preferred coordinate labels.

4.2 Three auxiliary bilinear forms

Write $P_\perp = \mathbf{1} - P$. The positive metric of [3] is

$$q(u,v) = \widehat{g}(P_\perp u, P_\perp v) + \widehat{g}(JPu, JPv). \tag{4.3}$$

Writing $u = t + n$, $t \in E, n \in F$, gives $q(u,u) = \widehat{g}(n,n) + \widehat{g}(Jt,Jt) > 0$ for $u \neq 0$. The two summands exchange under J. In the restricted ansatz assigning coefficients a, b to these two terms, J-invariance requires $a = b > 0$. This fixes a relative coefficient within that ansatz, not every possible natural positive norm once additional fields are permitted. It obeys $q(Ju,Jv) = q(u,v)$ and $q(Ku,Kv) = q(u,v)$. Define

$$\eta_{\mathrm{B}}(u,v) = q(u,Lv), \qquad \omega(u,v) = q(u,Jv). \tag{4.4}$$

The subscript on η_{B} distinguishes this auxiliary pairing from the scalar potential parameter η.

Proposition 4.1 (Induced Born structure)**.** *The form η_{B} is symmetric of signature $(4,4)$, ω is a nondegenerate two-form, and q is positive definite. As matrices they satisfy*

$$\eta_{\mathrm{B}}^{-1} q = q^{-1}\eta_{\mathrm{B}} = L, \qquad \omega^{-1} q = -q^{-1}\omega = -J, \qquad \eta_{\mathrm{B}}^{-1}\omega = K. \tag{4.5}$$

Both E and F are maximally null for η_{B} and maximally isotropic for ω.

Proof. The q-adjoints of J, K, L are $-J, K, L$, respectively. This proves the symmetry and skew symmetry in (4.4); invertibility follows from (4.2). Since L exchanges E and F, η_{B} vanishes on either separately and is nondegenerate between them. It therefore has neutral signature. The same argument gives isotropy for ω. The inverse identities follow from $L^2 = \mathbf{1}$, $J^2 = -\mathbf{1}$, and $LJ = K$. □

These are the compatibility relations defining an auxiliary Born structure in the conventions of [18]. No assertion that $\mathrm{d}\omega = 0$ is included: until closure is imposed, ω is almost symplectic. The construction is a direct consequence of the CR geometric variables. It does not import a string action or identify CR with double field theory.

4.3 The missing datum is the Lorentzian clock metric

Neither q nor η_{B} is the physical eight-dimensional metric. Their signatures alone rule out those identifications. The distinction can be stated constructively.

On E, define the q-self-adjoint endomorphism A by

$$g_E(u,v) = q(u, \mathsf{A}v), \qquad u, v \in E, \tag{4.6}$$

where $g_E = \widehat{g}|_E$. It has one negative and three positive eigenvalues. The complete physical metric is recovered by

$$\widehat{g}(u,v) = q(Pu, \mathsf{A}Pv) + q(P_\perp u, P_\perp v). \tag{4.7}$$

Conversely, a Born triple with the polarization K, together with such an A, reconstructs the original algebraic data and the positive metric (4.3). This is an invertible description of the stated signature branch.

Locally, a particularly simple normalized branch is

$$\mathsf{A}^2 = \mathbf{1}, \qquad \mathsf{A} = \mathbf{1}_E - 2\tau \otimes \tau_q^\flat, \qquad q(\tau,\tau) = 1, \quad \tau \in E. \tag{4.8}$$

Then

$$\widehat{g} = q - 2\tau_q^\flat \otimes \tau_q^\flat. \tag{4.9}$$

A positive geometry becomes one-time Lorentzian by reversing the sign of one clock line. The rank-one projector $\tau \otimes \tau_q^\flat$ is independent of the sign of τ. A globally chosen unit vector τ additionally requires orientation of that timelike line; on a time-oriented bulk this can be chosen consistently. The normalized flat bulk and flat-seed defect have this form. The condition $\mathsf{A}^2 = \mathbf{1}$ is a restriction, not a consequence of P1 or P2. In general its four eigenvalues are additional data. Whether the distinguished clock line has observable effects depends on the action and matter couplings; the decomposition alone does not settle that question.

Thus a useful description of the parent geometry is *an auxiliary Born structure supplied with Lorentzian clock data and scalar maps.* It is more informative than calling the whole object simply a complex metric.

5 The signature problem for an eight-dimensional metric

To give the four normal directions independent physical data, it is tempting to reinterpret Γ directly as a real eight-dimensional metric. For a complex bilinear metric, this step has an unavoidable signature consequence.

At a point write $\Gamma = A + \mathrm{i}B$ with real symmetric matrices A and B. On real tangent coordinates (v, w),

$$\mathcal{R} = \mathrm{Re}[\Gamma(\mathrm{d}z, \mathrm{d}z)] \quad \longleftrightarrow \quad \begin{pmatrix} A & -B \\ -B & -A \end{pmatrix}. \tag{5.1}$$

Here $\mathcal{R}$ denotes realification, not the positive Born metric q. The underlying complex structure satisfies

$$J = \begin{pmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad J^{\mathsf{T}}\mathcal{R}J = -\mathcal{R}. \tag{5.2}$$

Hence J exchanges positive and negative directions. Nondegeneracy forces the real signature to be $(4,4)$, irrespective of the Lorentzian signature measured on M. In the simple case $\Gamma = (1 + \mathrm{i}\chi)\eta$,

$$\mathcal{R}^2 = (1+\chi^2)\mathbf{1}_8, \qquad \mathrm{tr}\,\mathcal{R} = 0, \tag{5.3}$$

so four eigenvalues have each sign. Replacing the bilinear metric by the natural Hermitian Lorentz form $\eta_{a\bar{b}}\mathrm{d}z^a\mathrm{d}\bar{z}^b$ instead gives real signature $(2,6)$. It still has two time directions.

At $\Gamma = \eta$, the realified bilinear wave equation has principal part

$$-\partial_t^2 + \sum_{i=1}^{3}\partial_{x_i}^2 + \partial_{y_0}^2 - \sum_{i=1}^{3}\partial_{y_i}^2. \tag{5.4}$$

For Fourier wave numbers $(\mathbf{k}, \kappa_0, \boldsymbol{\kappa})$,

$$\omega^2 = |\mathbf{k}|^2 + \kappa_0^2 - |\boldsymbol{\kappa}|^2. \tag{5.5}$$

Unrestricted codimension-one data include modes with arbitrarily fast exponential growth in the selected time t. The usual one-time hyperbolic initial-value formulation and its positive kinetic interpretation therefore do not follow from this realification.

This is a specific obstruction, not a theorem forbidding all multiple-time models. Craig and Weinstein construct well-posed ultrahyperbolic Cauchy problems under nonlocal restrictions on the allowed data [9]. Such restrictions would be additional physical structure; P1 and P2 as stated do not supply them. The physical metric $\widehat{g}$ used throughout this article is instead an independent real metric of signature $(1, 7)$.

5.1 The physical one-time metric cannot be Hermitian

Suppose $\widehat{g}(Ju, Jv) = \widehat{g}(u, v)$. If u is timelike, then Ju is equally timelike and $\widehat{g}(u, Ju) = 0$. The two independent vectors span a negative-definite plane, contradicting a metric with only one negative direction. Thus a nondegenerate J-Hermitian real metric has even timelike index.

This excludes identifying the one-time bulk metric with a Hermitian metric. It does not exclude a complex structure on the manifold. The positive auxiliary q in (4.3) is Hermitian; the physical $\widehat{g}$ is not. Nor may the physical metric simply be identified with the realification of the complex-bilinear Γ, which has split signature in the adapted CR construction.

6 The complex clock readout

For a real scalar χ, define the complex-linear map into $E \otimes \mathbb{C}$

$$Z = P - \mathrm{i}PJ, \qquad ZJ = \mathrm{i}Z, \tag{6.1}$$

and the complex symmetric form

$$\Gamma(u, v) = (1 + \mathrm{i}\chi)\widehat{g}(Zu, Zv), \tag{6.2}$$

where $\widehat{g}$ is extended complex bilinearly.

Proposition 6.1 (Algebraic complex extension)**.** *For fixed $(\widehat{g}, J, P, \chi)$, (6.2) is the unique complex-bilinear form relative to J whose restriction to E is $(1 + \mathrm{i}\chi)\widehat{g}|_E$. It is nondegenerate as a metric on the complex rank-four tangent bundle. On a regular core its real light cone is exactly that of the induced clock metric.*

Proof. The identity $ZJ = \mathrm{i}Z$ gives $\Gamma(Ju, v) = \Gamma(u, Jv) = \mathrm{i}\Gamma(u, v)$. For $u \in E$, $Pu = u$ and $PJu = 0$, so the stated restriction holds. Every real tangent is a sum of a vector in E and one in JE; complex bilinearity therefore determines all remaining components uniquely. Nondegeneracy follows from that of $\widehat{g}|_E$ and $1 + \mathrm{i}\chi \neq 0$. A real tangent to the core has zero complex interval precisely when its real clock interval is zero. □

In an eight-by-eight complex matrix acting on the complexification of the real tangent space, a $(2, 0)$ form has complex rank four and annihilates $T^{0,1}$. This is the expected type decomposition, not degeneracy of the rank-four complex metric.

For a real covector on the core, the shell prescription

$$\mathcal{Q}_{\mathbb{C}} = (1 + \mathrm{i}\chi)^{-1}(g^{\mu\nu}p_\mu p_\nu + m^2) \tag{6.3}$$

realizes P2. The intrinsic real particle Hamiltonian provides a real flow on the core. This prescription does not prove that unrestricted bulk geodesics remain on an arbitrarily bent core, or derive a localized matter action. Those are separate dynamical claims.

7 Holomorphic geometry and a real structure

7.1 A local complex neighborhood

Let $\mathcal{Z}$ be a complex four-manifold and σ an antiholomorphic involution on a neighborhood $\mathcal{U}$. Choose a four-dimensional connected component M of its fixed locus. In adapted local coordinates,

$$z^a = x^a + \mathrm{i}y^a, \qquad \sigma(z) = \bar{z}, \qquad M: \ y = 0. \tag{7.1}$$

The slice is maximally real. The analytic category matters: a real-analytic manifold admits local complexification, and a maximally real analytic embedding can be straightened locally. General facts about such complexification germs are recalled, for example, in the preliminaries of Greb and Wong [6]. No Kähler condition is assumed here. A global complexification with the required topology is a further problem.

For a holomorphic tensor define its reflected conjugate by

$$T^\sharp = \sigma^* \overline{T}. \tag{7.2}$$

This operation is antilinear, maps holomorphic tensors to holomorphic tensors, and satisfies $(T^\sharp)^\sharp = T$. For a scalar in adapted coordinates, $T^\sharp(z) = \overline{T(\bar{z})}$. It is not pointwise conjugation at the same complex coordinate.

For a nondegenerate holomorphic complex symmetric metric Γ_+, put $\Gamma_- = {\Gamma_+}^\sharp$ and introduce

$$H = \frac{\Gamma_+ + \Gamma_-}{2}, \qquad B = \frac{\Gamma_+ - \Gamma_-}{2\mathrm{i}}. \tag{7.3}$$

Both tensors are sharp-real: $H^\sharp = H$ and $B^\sharp = B$. Their restrictions to M are real; we require $g = H|_M$ to be Lorentzian. In a sufficiently small neighborhood of a regular slice point, H is also nondegenerate.

The required reality condition is on H, B, and the appropriate scalar fields. It is *not* $\Gamma_- = \Gamma_+$. Imposing the latter would set $B = 0$ and remove the CR scalar. In the constrained theory,

$$B = \Xi H, \qquad \Xi^\sharp = \Xi, \qquad \Gamma_+ = F_+ H, \quad \Gamma_- = F_- H, \quad F_\pm = 1 \pm \mathrm{i}\Xi. \tag{7.4}$$

On M, $\Xi = \chi$ and (1.2) follows. Away from M, even sharp-real tensors generally have complex-valued components.

We write $\Gamma_+ = H + \mathrm{i}B$ and $\Gamma_- = H - \mathrm{i}B = {\Gamma_+}^\sharp$. The two tensors are reflected partners, not two independently propagating real gravities. The holomorphic clock extension H, physical bulk metric $\widehat{g}$, and positive auxiliary metric q remain distinct. The imaginary holomorphic tensor B_{ab} is not the bulk construction B. The algebraic readout Γ and holomorphic Γ_+ agree on their prescribed real trace when the conditions below hold; no off-slice identity between them is assumed.

7.2 The restriction that holomorphicity imposes

Theorem 7.1 (Uniqueness of the holomorphic germ)**.** *A holomorphic tensor that vanishes on an open subset of a maximally real analytic slice vanishes in a neighborhood of that subset. Consequently, the complete real-analytic slice values of a holomorphic tensor determine its germ uniquely.*

Proof. Straighten the slice as in (7.1) and consider one holomorphic component $t(z)$. Since $t(x) = 0$ on an open real set, every real coordinate derivative of this restriction vanishes. Holomorphicity identifies these derivatives with the corresponding complex Taylor coefficients. The

convergent holomorphic Taylor series is zero. Apply this argument component by component and cover the slice patch. □

In particular,

$$\partial_{y^a} T = \mathrm{i}\partial_{x^a} T, \qquad \partial_y^\alpha T\Big|_M = \mathrm{i}^{|\alpha|}\partial_x^\alpha (T|_M), \tag{7.5}$$

for tensor components in adapted holomorphic coordinates. These are coordinate derivatives, not an additional covariant transport law. Within the convergence domain,

$$T(x+\mathrm{i}y) = \sum_\alpha \frac{(\mathrm{i}y)^\alpha}{\alpha!}\partial_x^\alpha T(x). \tag{7.6}$$

Apply the theorem to $B - \Xi H$. If P1 holds on the entire real slice patch and the fields are holomorphic, the same proportionality holds throughout the germ. There are no additional holomorphic traceless components that vanish on M and become arbitrary just off it.

This statement concerns data on a *four-dimensional spacetime region*, including its time dependence. It does not say that positions alone on a three-dimensional Cauchy hypersurface determine a solution. Nor does it apply to arbitrary smooth nonholomorphic bulk fields. It explains precisely why an analytic extension and a bulk theory with independent normal initial data are different constructions.

8 Integrability, connections, and the holomorphic sector

8.1 Three differential conditions

The following conditions have different meanings:

$$\mathcal{N}_J = 0, \qquad \mathcal{N}_K = 0, \qquad \mathrm{d}\omega = 0. \tag{8.1}$$

The first gives complex coordinates under the usual smoothness assumptions. The second says that both E and F are involutive. The third makes ω symplectic. The alignment equations (3.4) make E involutive where $\mathrm{d}X^A$ have rank four: its leaves are their joint level sets. They do not force F to be involutive. Section 36 gives an explicit counterexample.

When both J and K are integrable, this is a complex-product geometry. In that setting there is a unique ordinary torsion-free connection preserving the pair (J, K) [19, 20]. It need not preserve $\widehat{g}$. Independently, a Born structure with its canonical D-bracket has a unique compatible connection with vanishing *generalized* torsion [18]. This latter condition is not vanishing ordinary torsion. These are conditional uniqueness statements for connections, not uniqueness statements for an action, potential, clock operator, or solution.

In the present construction $q|_E$ is positive, while g_E is Lorentzian. The Born connection theorem cannot therefore be used to identify its projected connection with the physical Lorentzian Levi-Civita connection without an additional relation involving A.

The three connections that will matter are therefore the Levi-Civita connection of $\widehat{g}$, the complex-product connection where both structures are integrable, and the Born-compatible connection with its generalized torsion prescription. Section 26 gives an explicit background on which the first is curved while the complex-product connection is flat. Connection uniqueness has a precise domain; it does not by itself remove the potential freedom demonstrated in Section 13.

Part III

Parent theory and variational dynamics

9 The specified four-dimensional CR realization

The parent sought here must recover the following particular CR realization:

$$S_4 = \int_M \mathrm{vol}_g \left[\frac{M_{\mathrm{P}}^2}{2} R[g] - \frac{1}{2} \mathsf{G}_{IJ}(\Phi) g^{\mu\nu} \partial_\mu \Phi^I \partial_\nu \Phi^J \right]. \tag{9.1}$$

There is no potential in the baseline construction. With n_{p} response pairs,

$$\Phi = (\phi, U_1, V_1, \ldots, U_{n_{\mathrm{p}}}, V_{n_{\mathrm{p}}}), \qquad f^2 = \frac{3}{2} M_{\mathrm{P}}^2, \qquad \chi = \sinh(\phi/f), \tag{9.2}$$

$$\mathrm{d}s_{\mathsf{G}}^2 = \mathrm{d}\phi^2 + \sum_{a=1}^{n_{\mathrm{p}}} \mathrm{d}q_a^{\mathsf{T}} \mathsf{K}_{C_a}(\phi/f) \mathrm{d}q_a, \qquad q_a = \begin{pmatrix} U_a \\ V_a \end{pmatrix}, \quad C_a > 1, \tag{9.3}$$

$$\mathsf{K}_C(\theta) = \begin{pmatrix} \dfrac{C\cosh\theta + 1}{C+1} & -\dfrac{\sinh\theta}{\sqrt{C^2-1}} \\ -\dfrac{\sinh\theta}{\sqrt{C^2-1}} & \dfrac{C\cosh\theta - 1}{C-1} \end{pmatrix}. \tag{9.4}$$

For real θ, the leading diagonal entry is positive and

$$\det \mathsf{K}_C = \cosh^2\theta > 0, \qquad \mathsf{K}_C(0) = \mathbf{1}. \tag{9.5}$$

The target is therefore positive definite. The response parameters are written with uppercase C; neither their integrality nor a discrete spectrum is assumed.

The coefficient of the geometric scalar follows from the chosen constrained real-part complex curvature action. The response metric is an additional dynamical choice compatible with the postulates. In particular, P1 and P2 do not already determine K_C. The companion constraint analysis proves closure after the regular algebraic reduction and exhibits positive target deformations that obey the same tests [5].

Dimensionless fields and constraint scope. For the bulk constructions it is useful to set

$$\vartheta = \phi/f, \qquad u_j = U_j/M_{\mathrm{P}}, \qquad v_j = V_j/M_{\mathrm{P}}, \qquad Q = (\vartheta, u_1, v_1, \ldots), \qquad n = 1 + 2n_{\mathrm{p}}. \tag{9.6}$$

Then $\mathsf{G}_{IJ}\mathrm{d}\Phi^I\mathrm{d}\Phi^J = M_{\mathrm{P}}^2 \mathfrak{h}_{IJ}\mathrm{d}Q^I\mathrm{d}Q^J$, where

$$\mathfrak{h}_{IJ}\mathrm{d}Q^I\mathrm{d}Q^J = \frac{3}{2}\mathrm{d}\vartheta^2 + \sum_{j=1}^{n_{\mathrm{p}}} \mathrm{d}w_j^{\mathsf{T}} \mathsf{K}_{C_j}(\vartheta) \mathrm{d}w_j, \qquad w_j = (u_j, v_j)^{\mathsf{T}}. \tag{9.7}$$

Equivalently,

$$S_4 = \frac{M_{\mathrm{P}}^2}{2} \int_M \mathrm{vol}_g \left[R[g] - \mathfrak{h}_{IJ} g^{\mu\nu} \partial_\mu Q^I \partial_\nu Q^J \right]. \tag{9.8}$$

Only ϕ is globally canonical in this coordinate convention. Each response pair is canonically normalized at $\vartheta = 0$; its target metric remains field dependent elsewhere.

After the regular algebraic variables have been eliminated, this is Einstein gravity coupled to n real scalars with a positive target. Its four spacetime-diffeomorphism constraints obey the usual hypersurface-deformation algebra, giving $2 + n = 3 + 2n_{\mathrm{p}}$ physical configuration degrees of freedom. The closure statement concerns this reduced classical system. It is not a complete raw Dirac classification of every multiplier and holomorphic variable later used to represent it. The bulk counts and their Hamiltonian constraints are given in Appendix B.

9.1 Why unconstrained complex vacuum gravity is insufficient

Variation within $\gamma = (1 + \mathrm{i}\chi)g$ is different from varying an arbitrary complex metric and imposing that formula afterward. A simple counterexample makes the distinction concrete. For $t > 0$,

$$g = -\mathrm{d}t^2 + (t/t_0)^{2/3}\mathrm{d}\mathbf{x}^2, \qquad \phi = \sqrt{\frac{2}{3}}M_{\mathrm{P}}\log(t/t_0) \tag{9.9}$$

solves (9.1) with constant response fields. At $t = t_0$, its induced complex metric satisfies

$$R_{tt}[\gamma] = \frac{2\mathrm{i}}{3t_0^2}, \qquad R_{ij}[\gamma] = \frac{2\mathrm{i}}{9t_0^2}\delta_{ij}. \tag{9.10}$$

Thus $R_{ab}[\Gamma_+] = 0$ for an otherwise unconstrained holomorphic metric cannot be the parent equations of every solution of this CR model. A suitable parent must retain the constrained variational structure or supply different bulk equations whose omitted components are actually satisfied.

10 A common scalar parent at fixed couplings

10.1 Why the field space is enlarged

The original defect field takes values in S^4, with $\mathcal{U}(N) = [24(N^5)^2 - 12]/\ell^2$. Freezing that field would recover a potential-free bulk only at a stationary point with $\mathcal{U} = 0$. Its stationary loci instead have values $-12/\ell^2$ and $+12/\ell^2$. Thus the old D theory alone does not contain B as a constant-field sector. A proper great equator $S^4 \subset S^5$ provides room for additional stationary zero-energy poles. Within the round-sphere family, S^5 is the smallest sphere with the required proper great S^4 equator. This is the reason for the new target, not an extra spacetime dimension or a claim of uniqueness among all possible enlargements.

10.2 Field content, action, and equations

Let $(\widehat{M}, \widehat{g})$ be an eight-dimensional real Lorentzian manifold with one time direction. Retain the dimensionless CR fields Q^I, $I = 1, \ldots, n$, and their positive target metric $\mathfrak{h}$. Its convention is fixed by (9.7). Introduce six dimensionless real scalars

$$X^\alpha X^\alpha = 1, \qquad \alpha = 1, \ldots, 6. \tag{10.1}$$

There are five independent new scalar degrees of freedom. Indices $A = 1, \ldots, 4$ and $a = 1, \ldots, 5$ denote subsets of these components. Define

$$R_5^2 := \sum_{a=1}^{5}(X^a)^2, \tag{10.2}$$

where R_5 is a target-coordinate abbreviation, not a spacetime radius. The potential is

$$\boxed{\mathcal{V}_\eta(X) = \frac{12}{\ell^2}\left[(X^5)^2 - \sum_{A=1}^{4}(X^A)^2 + \eta(X^6)^2R_5^2\right].} \tag{10.3}$$

We use the displayed polynomial also off the unit sphere when varying the multiplier action. Different off-sphere continuations shift the multiplier; specifying this convention avoids an apparent mismatch later. The dimensionless parameter η is fixed once and for all. Choosing any $\eta > 1$, for example $\eta = 2$, satisfies the additional scalar tests below.

The common action is

$$S_*[\widehat{g}, Q, X, \Lambda] = \frac{M_8^6}{2} \int_{\widehat{M}} \mathrm{vol}_{\widehat{g}} \Big[\widehat{R} - \mathfrak{h}_{IJ}(Q) \partial_M Q^I \partial^M Q^J - 6 \partial_M X^\alpha \partial^M X^\alpha - 2\mathcal{V}_\eta(X) + \Lambda(X^\alpha X^\alpha - 1) \Big]. \tag{10.4}$$

The target of the new fields is the round sphere with metric $6\mathrm{d}\Omega_5^2$. It is positive definite. M_8 has mass dimension one, while $\mathcal{V}_\eta$ and Λ have dimension length^{-2}. No expansion in energy, curvature, or field amplitude defines (10.4).

The trace-reversed Einstein and scalar equations are

$$\widehat{R}_{MN} = \mathfrak{h}_{IJ} \partial_M Q^I \partial_N Q^J + 6 \partial_M X^\alpha \partial_N X^\alpha + \frac{\mathcal{V}_\eta}{3} \widehat{g}_{MN}, \tag{10.5}$$

$$0 = \widehat{\Box} Q^I + \Gamma^I{}_{JK}(\mathfrak{h}) \widehat{g}^{MN} \partial_M Q^J \partial_N Q^K, \tag{10.6}$$

$$0 = 6 \widehat{\Box} X^\alpha - \mathcal{V}_{\eta,\alpha} + \Lambda X^\alpha, \tag{10.7}$$

$$\Lambda = 6(\partial X)^2 + X^\alpha \mathcal{V}_{\eta,\alpha}. \tag{10.8}$$

The last relation follows by contracting (10.7) with X. The unit constraint makes the multiplier contribution to the on-shell stress vanish. The coefficient $1/3$ in (10.5) is $2/(8-2)$.

10.3 What is common, and what remains additional

At fixed (M_8, ℓ, η, C_j), the constant poles $X^a = 0, X^6 = \pm 1$ recover the entire potential-free B theory. The invariant equator $X^6 = 0$ recovers the entire D theory, including its discarded scalar equation and a shift of the unit-constraint multiplier. Sections 14 and 18 give the full proofs. These are restrictions of one action rather than coupling limits. The common scalar parent has $20 + n + 5 = 25 + n$ local physical configuration degrees of freedom after the sphere constraint is eliminated.

The action supplies neither a kinetic equation for J nor a rule selecting P in every solution. Its pole sector has no regular order-parameter core: all four selecting gradients vanish there. At a general regular zero of $X^1, \ldots, X^4$, the remaining unit constraint allows a circle in the (X^5, X^6) plane. The old north-core condition is a specified sector of that larger field space.

The exact restrictions concern the equations on a chosen manifold. The compact torus and noncompact defect subsequently use different global domains and boundary conditions. The common local action does not prove a dynamical change of spacetime topology between them. It also does not fix η or the response parameters. The strict range $\eta > 1$ is justified by the transverse scalar calculations below.

11 Auxiliary holomorphic matching on a regular slice

Let M be a real-analytic maximally real slice, with Lorentzian $g = \iota^* \widehat{g}$ and put $\chi = \sinh(\vartheta|_M)$. Introduce a holomorphic complex symmetric metric Γ_+ on a neighborhood of M and a complex symmetric slice multiplier $\mathcal{M}^{\mu\nu}$. Add

$$S_{\mathrm{match}} = \mathrm{Re} \int_M \mathrm{vol}_g \, \mathcal{M}^{\mu\nu} \left[(\iota^* \Gamma_+)_{\mu\nu} - (1 + \mathrm{i} \sinh \vartheta) g_{\mu\nu} \right], \tag{11.1}$$

$$S_{\mathrm{read}} = S_{\mathrm{bulk}} + S_{\mathrm{match}}. \tag{11.2}$$

The traces of Γ_+ admit independent real and imaginary analytic variations. Variation of these components gives $\mathcal{M} = 0$, while variation of $\mathcal{M}$ gives

$$\iota^*\Gamma_+ = (1 + \mathrm{i}\sinh\vartheta)g. \tag{11.3}$$

Consequently all localized terms in the $\widehat{g}$ and Q equations from (11.1) vanish on shell. Here S_{bulk} may be S_*, B, or D on the corresponding regular sector. The multiplier is not a tension or a hidden source supporting the slice. For analytic bulk traces, Theorem 7.1 supplies a unique local holomorphic Γ_+, after shrinking the neighborhood to preserve nondegeneracy.

This construction retains a holomorphic CR metric near M, while allowing the real bulk fields to depend independently on the four normal coordinates. Its cost is precise: Γ_+ is an *auxiliary holomorphic readout* of the slice geometry. Its analytic continuation need not equal any simple combination of $\widehat{g}(x, y)$ and $Q(x, y)$ away from M, and does not determine their kinetic equations. We have not shown that this germ extends around the entire internal torus. Matching therefore supplies a common readout prescription, not an independent kinetic equation for the holomorphic metric.

The same reasoning applies when the integration surface is the regular field-defined core M_X: every term from varying its position or its induced volume is proportional to $\mathcal{M}$ and vanishes after $\mathcal{M} = 0$. This argument assumes the analytic core branch and the independent analytic trace variations discussed in Appendix A. It does not add a localized gravitational action or matter tension. The tensor reaction in H is different and is retained in Section 27.

12 An explicit joint trial action

The preceding variables can be placed in one variational system. To make this statement concrete, for an endomorphism T define

$$\mathfrak{K}[T] = \widehat{g}^{MN} q_{RS} q^{UV} (\widehat{\nabla}_M T^R{}_U)(\widehat{\nabla}_N T^S{}_V), \tag{12.1}$$

where $\widehat{\nabla}$ is the Levi-Civita connection of $\widehat{g}$. A trial extension is

$$S_{\mathrm{trial}} = S_* + \frac{M_8^6}{2} \int \mathrm{vol}_{\widehat{g}} \left[-\frac{\kappa_J}{2}\mathfrak{K}[J] - \frac{\kappa_P}{2}\mathfrak{K}[P] + \mathcal{L}_{\mathrm{con}} \right]. \tag{12.2}$$

The multiplier density can be specified componentwise as

$$\begin{aligned} \mathcal{L}_{\mathrm{con}} = {} & \mathcal{A}_M{}^N (J^M{}_R J^R{}_N + \delta^M_N) + \mathcal{B}_M{}^N (P^M{}_R P^R{}_N - P^M{}_N) \\ & + \zeta_0 (P^M{}_M - 4) + \mathcal{C}^{MN} (\widehat{g}_{MR} P^R{}_N - \widehat{g}_{NR} P^R{}_M) \\ & + \mathcal{D}_M{}^N (P^M{}_R J^R{}_N + J^M{}_R P^R{}_N - J^M{}_N) \\ & + \mathcal{E}_A{}^N P^M{}_N \partial_M X^A + \mathcal{T}_K{}^{MN} (\mathcal{N}_J)^K{}_{MN}. \end{aligned} \tag{12.3}$$

The last multiplier is antisymmetric in M, N. Algebraic multipliers have dimension length^{-2}; those multiplying one derivative have dimension length^{-1}. Redundancies among the displayed constraints and their multiplier symmetries must be resolved in a canonical analysis.

Here $\widehat{g}, Q, X, J, P$ and every multiplier are varied. In particular, q and $\widehat{\nabla}$ in (12.1) are field dependent; their variations cannot be discarded. The X equations now include alignment reaction terms, the J and P equations include derivatives, and the metric equation includes their stress. The slice, complex structure, and real geometry therefore participate in one action rather than being assigned after solving the metric equations.

This is an explicit variational *framework*, not an established healthy parent. For fixed background geometry, positive q and positive κ_J, κ_P give a positive fiber norm for the time

derivatives of these tensor fields. They do not settle the full problem: $\widehat{\nabla} J$ and $\widehat{\nabla} P$ mix metric derivatives with the new fields; $\mathcal{N}_J = 0$ is a derivative constraint; alignment loses rank on the constant-pole sector. A Dirac–Bergmann analysis and a physical principal-symbol calculation are still required. Counting unconstrained tensor components would not supply the physical degrees of freedom or prove a common characteristic cone.

A flat configuration with constant CR fields, pole X, standard constant J, constant P, and all extra multipliers zero is a solution of this trial system. That limited existence check does not imply that the old nontrivial bulk or defect solutions survive. The tests in Section 26 show why this does not extend automatically to the old warped defect.

13 What the current principles do not uniquely select

13.1 A deformation invisible to the sector and linear tests

For any real ε, replace the potential by

$$\mathcal{V}_{\eta,\varepsilon} = \mathcal{V}_\eta + \delta\mathcal{V}_\varepsilon, \qquad \delta\mathcal{V}_\varepsilon = \frac{\varepsilon}{\ell^2}(X^6)^4(R_5^2)^2. \tag{13.1}$$

This is a smooth, bounded potential on the compact target. It preserves the rotations of the first four components and the sign symmetries of components five and six. It adds no derivatives and changes no scalar kinetic signature.

Proposition 13.1 (Nonuniqueness under the presently stated tests)**.** *The value, first derivatives, and second derivatives of $\delta\mathcal{V}_\varepsilon$ vanish everywhere on the equator and at both poles. Consequently the complete B and D restrictions, their stress tensors and field equations, and all quadratic variations about solutions within those restrictions are unchanged. Their existing P1–P2 realization and conditional H reconstruction are also unchanged. Nevertheless the unrestricted nonlinear theories differ.*

Proof. At the equator the factor $(X^6)^4$ makes the first two jets vanish. At either pole, $(R_5^2)^2$ is fourth order in the five transverse Cartesian target coordinates. The potential contributes neither background stress nor first-order equations nor quadratic terms on these loci. The geometric and particle prescriptions realizing P1 and P2 have not been altered.

To see the nonlinear difference in a regular pole chart, set $X = (p^a, \pm\sqrt{1-p^2})$. Then

$$\delta\mathcal{V}_\varepsilon = \frac{\varepsilon}{\ell^2}(p^2)^2(1-p^2)^2 = \frac{\varepsilon}{\ell^2}(p^2)^2 + O(|p|^6). \tag{13.2}$$

With canonically normalized eight-dimensional scalars $\varphi^a = \sqrt{6}M_8^3 p^a$, the added quartic Lagrangian is

$$\delta\mathcal{L}^{(4)} = -\frac{\varepsilon}{36M_8^6\ell^2}(\varphi^a\varphi^a)^2. \tag{13.3}$$

It changes nonlinear interactions while preserving the quadratic masses. It is not a term proportional to the unit constraint or a spacetime boundary term. □

One can also track a critical potential value. Along $X^5 = 0$, write $t = (X^6)^2$. For $\eta > 1$, the old interior barrier is at $t_* = (\eta+1)/(2\eta)$, with value $3(\eta-1)^2/(\eta\ell^2)$. Its first-order change under (13.1) is $\varepsilon t_*^2(1-t_*)^2/\ell^2$, nonzero. At $\eta = 2$, it is $9\varepsilon/(256\ell^2)$. Such critical values of the potential on the target do not depend on its coordinate parametrization.

This proposition does not assert that every deformed theory has been proved globally stable. No such theorem exists for the undeformed defect either. It proves that P1, P2, the established exact restrictions, and their linear checks do not select the nonlinear completion. Replacing

the constant ε by suitable smooth invariant functions yields still more freedom. A deeper principle could exclude these terms, but it must actually do so rather than be inferred from the checks they all pass.

13.2 The action class must be specified

The goal of a unique mother theory can be formulated as a classification problem. One must declare the field content, gauge symmetries, regularity, spacetime signature, allowed derivative order, and equivalences of actions. Total divergences and invertible field redefinitions are conventional equivalences; changes in physical nonlinear interactions are not. One must also state whether dimensional and dimensionless constants are to be predicted or left as parameters. A unique action does not imply a unique solution or uniquely selected initial data.

Several plausible restrictions do less than their names suggest:

Requirement	What it does not yet determine
P1 and P2	Bulk interactions invisible to the slice cone and shell
Exact B/D sectors and quadratic spectra	The deformation (13.1)
Fixed field content and positive scalar target	Non-derivative invariant potentials
Complex integrability	A metric action, a selected real core, or holomorphicity of every field
Positive tensor fiber norms	Constraint closure and the coupled metric kinetic matrix
Second-order metric equations	A unique eight-dimensional gravity action

The last point has a standard dimensional qualification. In eight dimensions, second-order metric field equations allow the cosmological, Einstein–Hilbert, quadratic Lovelock, and cubic Lovelock terms. The fourth Lovelock density is the eight-dimensional Euler density and does not alter bulk local equations under the usual fixed-topology variational assumptions. For example,

$$\mathcal{L}_2 = R_{MNPQ}R^{MNPQ} - 4R_{MN}R^{MN} + R^2 \tag{13.4}$$

is not a topological density in eight dimensions. Lovelock's classification therefore does not reduce to the four-dimensional Einstein conclusion here [17]. A restriction to a two-derivative *Lagrangian* is stronger than requiring second-order *equations*. Even the stronger restriction does not remove the zero-derivative ambiguity (13.1). No claim is made that arbitrary Lovelock terms preserve the old defect or all desired characteristic cones.

A productive additional principle must make a definite prediction in this problem. It could, for instance, relate the scalar potential to a specified geometric curvature or connection invariant and thereby fix the allowed functions. But the underlying geometry, connection, and relation would be new assumptions until derived. A general instruction to use "minimal", "natural", or "stable" geometry does not by itself exclude (13.1) or fix η.

These are criteria for an exact fundamental variational theory. No obstruction above depends on interpreting CR as an effective low-energy expansion. The final section states the remaining calculations in terms of the geometric variables already defined.

Part IV

B, D, and H

14 B: the pole sector and independent bulk dynamics

The first exact restriction removes the order parameter's stress without changing the remaining bulk dynamics. Independent dependence on all four normal coordinates remains allowed; the flat torus is a further solution ansatz, not the definition of B.

14.1 The two poles recover B

Set

$$X^1 = \cdots = X^5 = 0, \qquad X^6 = \pm 1. \tag{14.1}$$

For the polynomial (10.3), both $\mathcal{V}_\eta$ and all six ambient first derivatives vanish at either pole. All field gradients vanish and $\Lambda = 0$. Hence the entire X stress is zero, and every equation in (10.7) is satisfied.

Proposition 14.1 (Pole reduction)**.** *At either pole* (14.1)*, S_* reduces consistently to*

$$S_B = \frac{M_8^6}{2} \int \mathrm{vol}_{\widehat{g}} \left[\widehat{R} - \mathfrak{h}_{IJ} \partial Q^I \partial Q^J \right]. \tag{14.2}$$

Every solution of B on its chosen spacetime lifts to a solution of S_* by the constant pole assignment. This statement is not limited to internally constant or flat-torus solutions.

Proof. The discarded equations and stress vanish as just shown. Equations (10.5) and (10.6) become exactly the B equations. The unit constraint is satisfied identically. Thus no further equation is imposed on $(\widehat{g}, Q)$. □

14.2 Geometry and field content

Work first on a real-analytic physical coordinate domain $M \subset \mathbb{R}^4$, with a neighborhood of the product

$$\widehat{\mathcal{Z}} = M \times T^4, \qquad y^A \sim y^A + 2\pi, \qquad R_A > 0. \tag{14.3}$$

Pair the four real coordinates with the four angles, writing $z^\mu = x^\mu + \mathrm{i} R_{\mu+1} y^{\mu+1}$ for $\mu = 0, \ldots, 3$. Imaginary translations supply the cylinder identifications and define an integrable complex structure J. A general bulk metric need not assign the reference radii R_A to these circles. The involution

$$\sigma : (x, y) \longmapsto (x, -y) \tag{14.4}$$

is antiholomorphic. Select the fixed component $y = 0$ as the physical slice. The torus has other fixed components; the bulk action does not select one of them. No orbifold quotient is required.

The dynamical bulk metric is a real $\widehat{g}_{MN}$ of signature $(1, 7)$, independent of the holomorphic CR metric. It is neither $\mathrm{Re}\,\Gamma_+$ nor a J-Hermitian metric. This is an added assumption, not a reinterpretation that evades the signature calculation. A general four-manifold need not admit this particular global complex cylinder structure; the construction proves a local parent and covers global product geometries when the stated complex structure exists.

Use the dimensionless fields Q and target $\mathfrak{h}$ of (9.6)–(9.7), now as fields on the real bulk. They may depend independently on all eight coordinates. Bulk covariance does not determine the factor $3/2$, the matrix function K_C, the number of response pairs, or their C_j. Those choices have been made to recover the specified CR slice dynamics.

14.3 The bulk action and equations

The bulk action is (14.2). The measure includes the metric determinant in angular coordinates; each radius contributes to the proper internal volume. M_8 has dimensions of mass. The scalar kinetic energy has the healthy sign because the target is positive and the spacetime has one time direction.

The Einstein equations and their trace-reversed form are

$$\widehat{R}_{MN} - \frac{1}{2}\widehat{g}_{MN}\widehat{R} = \mathfrak{h}_{IJ}\partial_M Q^I \partial_N Q^J - \frac{1}{2}\widehat{g}_{MN}\mathfrak{h}_{IJ}\partial_P Q^I \partial^P Q^J, \tag{14.5}$$

$$\widehat{R}_{MN} = \mathfrak{h}_{IJ}\partial_M Q^I \partial_N Q^J. \tag{14.6}$$

The scalar equations are

$$\widehat{\Box} Q^I + \Gamma^I{}_{JK}(\mathfrak{h})\widehat{g}^{MN}\partial_M Q^J \partial_N Q^K = 0. \tag{14.7}$$

In harmonic gauge these are quasilinear wave equations with principal metric $\widehat{g}$. The usual local hyperbolic construction applies on the Lorentzian branch with positive target [10]. For the holomorphic matching of Section 11 we restrict to real-analytic data and their local analytic development. Smooth bulk evolution by itself does not guarantee a holomorphic CR readout.

Neither J nor the reflection appears in the kinetic terms of (14.2). They supply the complex interpretation and the selected real slice. This separation is part of the model's present limitation: the bulk propagating geometry has not been derived from Γ_+ alone.

For analytic traces on the selected slice, use the matching prescription of Section 11. Its on-shell multiplier is zero and the B equations remain unchanged. The independent normal data live in $\widehat{g}, Q$, not in extra holomorphic jets that vanish on an entire real-slice spacetime patch.

15 Reflection preserves the physical slice

Restrict to solutions for which σ is an isometry of $\widehat{g}$ and the scalar fields are even. Locally,

$$\begin{aligned} \widehat{g}_{\mu\nu}(x,-y) &= \widehat{g}_{\mu\nu}(x,y), & \widehat{g}_{AB}(x,-y) &= \widehat{g}_{AB}(x,y), \\ \widehat{g}_{\mu A}(x,-y) &= -\widehat{g}_{\mu A}(x,y), & Q(x,-y) &= Q(x,y). \end{aligned} \tag{15.1}$$

Choose a sector in which M is Lorentzian and its four normal directions are spacelike. At $y = 0$,

$$\widehat{g}_{\mu A} = 0, \qquad \partial_A \widehat{g}_{\mu\nu} = 0, \qquad \partial_A Q = 0. \tag{15.2}$$

Tangential derivatives of the identically vanishing mixed trace also vanish. It follows that

$$\widehat{\Gamma}^A{}_{\mu\nu}\big|_M = 0. \tag{15.3}$$

Proposition 15.1 (Invariant real slice)**.** *The chosen nondegenerate fixed component is totally geodesic. A real bulk free probe initially tangent to it stays on it for as long as its geodesic exists in the neighborhood. Reflection-invariant constrained initial data, evolved in a reflection-compatible harmonic gauge, retain the symmetry under local uniqueness.*

Proof. Equation (15.3) is the vanishing normal second fundamental form. Equivalently, reflection maps a geodesic with initial point and tangent in the fixed locus to a geodesic with identical initial data. Uniqueness makes the two curves coincide. For the fields, the bulk equations are covariant under σ; the reflected solution has the same invariant initial data and hence agrees with the original development in a compatible gauge. □

With normal momentum zero, the bulk free shell reduces to

$$\widehat{g}^{MN}P_MP_N + m^2|_{M,\,P_A=0} = g^{\mu\nu}p_\mu p_\nu + m^2. \tag{15.4}$$

Matching then gives the complex representative (1.3). P1 is enforced by (11.3); P2 is preserved by the reflection-invariant real geodesic flow.

Total geodesy is not confinement. A probe given a nonzero normal velocity is allowed to leave the slice. A small displacement need not experience any restoring force. Likewise, this reflection argument neither localizes all interacting matter to M nor identifies the material clocks needed by an actual observer. It proves preservation of a specified classical sector, leaving its selection and physical robustness open.

16 An exact consistent reduction to four-dimensional CR

16.1 The reduction ansatz

Take a fixed flat internal torus and fields with no internal dependence:

$$\mathrm{d}\widehat{s}^2 = g_{\mu\nu}(x)\mathrm{d}x^\mu\mathrm{d}x^\nu + \sum_{A=1}^{4} R_A^2(\mathrm{d}y^A)^2, \qquad Q^I = Q^I(x). \tag{16.1}$$

The radii are positive constants. Define

$$\mathcal{V}_4 = (2\pi)^4\prod_{A=1}^{4} R_A, \qquad M_{\mathrm{P}}^2 = M_8^6\mathcal{V}_4, \tag{16.2}$$

$$\phi = f\vartheta, \qquad U_a = M_{\mathrm{P}}u_a, \qquad V_a = M_{\mathrm{P}}v_a, \qquad f^2 = \frac{3}{2}M_{\mathrm{P}}^2. \tag{16.3}$$

These relations give the correct mass dimension and canonical normalization of the four-dimensional fields. Substitution into (14.2) and integration over T^4 gives (9.1). By itself, this substitution is only an action calculation. A consistent reduction must also satisfy the equations of the omitted fields [11].

Theorem 16.1 (Exact potential-free bulk embedding)**.** *For every solution (g,Φ) of (9.1) with target (9.3), the configuration (16.1) with field map (16.3) solves all equations of (14.2). If the four-dimensional solution is real analytic, it also determines a local holomorphic metric and a solution of (11.2) with $\mathcal{M}=0$. Conversely, restriction of a bulk solution within this ansatz gives the same four-dimensional model.*

Proof. For the direct product with constant flat internal metric,

$$\widehat{R}_{\mu\nu} = R_{\mu\nu}[g], \qquad \widehat{R}_{\mu A} = 0, \qquad \widehat{R}_{AB} = 0. \tag{16.4}$$

Since $\partial_A Q^I = 0$, the right-hand side of (14.6) also vanishes in its mixed and internal components. The remaining equations are

$$R_{\mu\nu}[g] = \mathfrak{h}_{IJ}\partial_\mu Q^I\partial_\nu Q^J = M_{\mathrm{P}}^{-2}\mathsf{G}_{IJ}\partial_\mu\Phi^I\partial_\nu\Phi^J, \tag{16.5}$$

which are the trace-reversed four-dimensional Einstein equations. Constant internal volume gives $\widehat{\Box} Q^I = \Box_g Q^I$, and the target connection terms are the same under the constant field rescalings. The scalar equations therefore agree as well. All discarded mixed and internal equations have been checked, so the reduction is consistent. For analytic traces, the matching argument of Section 14 supplies the holomorphic readout without changing these equations. □

No condition such as $ER_A \ll 1$ was used. This is an exact subset of the classical bulk solution space, for arbitrary fixed radii. It is not a claim that generic bulk solutions are four-dimensional, nor that normal modes can be neglected in every experiment. The four-dimensional sector is not being obtained here as a derivative or low-energy expansion.

16.2 The internal Einstein equations really are nontrivial

The trace-reversed form makes consistency transparent, but can conceal why an apparent internal stress does not spoil it. Define

$$X = \mathfrak{h}_{IJ} g^{\mu\nu} \partial_\mu Q^I \partial_\nu Q^J. \tag{16.6}$$

On the product solution, the trace equation gives $R[g] = X$. In the un-reversed internal equation,

$$\widehat{\mathrm{Ein}}_{AB} = -\frac{1}{2}\widehat{g}_{AB} R[g], \qquad \widehat{T}_{AB} = -\frac{1}{2}\widehat{g}_{AB} X. \tag{16.7}$$

These terms are generally nonzero; they agree because of the external trace equation. Setting $\widehat{T}_{AB} = 0$ by hand would be an incorrect justification of the truncation.

16.3 An explicit lifted cosmology

The solution (9.9) lifts to

$$\begin{aligned} \mathrm{d}\widehat{s}^2 &= -\mathrm{d}t^2 + (t/t_0)^{2/3} \mathrm{d}\mathbf{x}^2 + \sum_A R_A^2 (\mathrm{d}y^A)^2, \\ \vartheta &= \frac{2}{3}\log(t/t_0), \qquad u_a = v_a = 0. \end{aligned} \tag{16.8}$$

Direct computation gives

$$\widehat{R}_{tt} = \frac{2}{3t^2} = \frac{3}{2}\dot{\vartheta}^2, \qquad \widehat{R}_{ij} = \widehat{R}_{AB} = \widehat{R}_{\mu A} = 0, \qquad \widehat{\Box}\vartheta = 0. \tag{16.9}$$

The scalar curvature is $-2/(3t^2)$. The nonzero internal Einstein tensor is $\widehat{g}_{AB}/(3t^2)$, precisely the internal scalar stress. This example is not a solution of unconstrained complex vacuum gravity, but is a solution of the explicitly defined bulk parent. The distinction is visible in the equations, rather than inferred from the resemblance of the metrics.

16.4 Why a generic potential spoils this simple embedding

Suppose one adds a real potential by replacing the bracket in (14.2) with

$$\widehat{R} - \mathfrak{h}_{IJ} \partial Q^I \partial Q^J - 2\mathcal{W}(Q). \tag{16.10}$$

In eight dimensions the trace-reversed equation becomes

$$\widehat{R}_{MN} = \mathfrak{h}_{IJ} \partial_M Q^I \partial_N Q^J + \frac{\mathcal{W}}{3}\widehat{g}_{MN}. \tag{16.11}$$

For a fixed flat torus with no internal scalar gradients, the internal equation requires $\mathcal{W} = 0$ along the solution. The scalar equations impose their additional potential derivatives in the

usual way. In the reduced action the dimensionful slice potential would be $V = M_{\mathrm{P}}^2 \mathcal{W}$. The symbol $\mathcal{U}$ denotes the distinct potential of the defect field. The theorem does not automatically extend to an arbitrary four-dimensional potential, a cosmological constant, or a proposed stabilizing term. Internal curvature, warping, flux, or additional fields require a new complete consistency calculation.

17 Degrees of freedom and normal modes

17.1 Bulk modes and four-dimensional zero modes

Before restricting to reflection-invariant configurations, an eight-dimensional massless graviton has

$$N_{\mathrm{grav}} = \left.\frac{D(D-3)}{2}\right|_{D=8} = 20 \tag{17.1}$$

local polarizations. The $n = 1 + 2n_{\mathrm{p}}$ real scalar fields add n positive kinetic modes on a regular background. Appendix B gives the corresponding constraint count for the bulk Einstein–sigma-model action.

Around the flat product vacuum, its unrestricted torus zero modes split into

Four-dimensional field	Multiplicity	Physical modes
Massless metric	1	2
Mixed metric vectors $\widehat{g}_{\mu A}$	4	8
Internal metric scalars $\widehat{g}_{AB}$	10	10
Total from the eight-dimensional graviton		20

The ten scalars include the internal volume and shape. Restriction to fields invariant under simultaneous $y \mapsto -y$ removes the constant mixed vectors, which are odd. It leaves the metric and all ten internal scalar zero modes: twelve gravitational zero-mode polarizations. Nonconstant odd mixed fields are not all removed.

The CR solution sector of Theorem 16.1 further sets the ten internal moduli to constants. This is consistent in the potential-free model, but they are not stabilized: perturbations changing them are genuine solutions of the larger theory. An exact truncation need not be an attractor or a dynamically preferred state.

17.2 The normal Fourier spectrum

Linearize around a constant-scalar flat product background. A normal Fourier mode has

$$\exp\left(\mathrm{i}\sum_A n_A y^A\right), \qquad n_A \in \mathbb{Z}, \tag{17.2}$$

and

$$-\Delta_{T^4} \longrightarrow m_{\mathbf{n}}^2 = \sum_A \frac{n_A^2}{R_A^2}, \qquad \omega^2 = |\mathbf{k}|^2 + m_{\mathbf{n}}^2. \tag{17.3}$$

For an additional nonnegative bulk mass, its square adds to $m_{\mathbf{n}}^2$. The normal contribution has the opposite behavior from the unstable transverse-time contribution in (5.5): the chosen bulk normal directions are spacelike. Reflection pairs the Fourier modes into the allowed cosine or sine combinations without changing these eigenvalues.

These are conditional linear spectra of the specified background. Their integers come from torus periodicity; they do not quantize the response parameter C. No observed particle has

yet been identified with a mode, no radii have been determined, and the torus vectors have not been shown to reproduce a non-Abelian gauge sector. In the reflection-invariant zero-mode sector, even the four Abelian mixed vectors are absent.

17.3 Causality and stability claims that are justified

The physical principal cone of the bulk equations is the $\widehat{g}$ light cone. Restricted to tangents of the physical slice, it is the g cone, and the matching complex metric has the same null tangents. Positive target metric and one timelike bulk direction exclude scalar kinetic ghosts in the classical model and give the usual local hyperbolic evolution on the regular branch. Equation (17.3) gives a nonnegative normal mass contribution about the flat product vacuum.

These results do not establish nonlinear global stability of every background, stability of compactification, absence of singularities, or a quantum positive Hilbert space. A nonflat target can support nontrivial background-dependent interactions even when its metric is positive. The unproven steps must be investigated for a specified background and a specified quantization, not inferred from the sign of one quadratic term.

The pole restriction is exact, but the full parent also has five fluctuations away from it. Their quadratic masses can be checked without freezing the metric moduli.

17.4 Parent fluctuations transverse to B

Near either pole use the regular chart

$$X = (p^1, \dots, p^5, \pm\sqrt{1 - p^a p^a}). \tag{17.4}$$

The target metric and potential are

$$6\mathrm{d}X \cdot \mathrm{d}X = 6\left(\delta_{ab} + \frac{p_a p_b}{1 - p^2}\right)\mathrm{d}p^a \mathrm{d}p^b, \tag{17.5}$$

$$\mathcal{V}_\eta = \frac{12}{\ell^2}\left[(\eta - 1)\sum_{A=1}^{4}(p^A)^2 + (\eta + 1)(p^5)^2 - \eta(p^2)^2\right]. \tag{17.6}$$

Thus the five scalar mass squares about a flat pole background are

$$m_A^2 = \frac{4(\eta - 1)}{\ell^2} \quad (A = 1, \dots, 4), \qquad m_5^2 = \frac{4(\eta + 1)}{\ell^2}. \tag{17.7}$$

The factor follows from the kinetic coefficient 6: $m^2 = \mathrm{Hess}\,\mathcal{V}_\eta/6$ at the origin of this chart. All five are positive for $\eta > 1$. The flat-torus geometry adds its nonnegative Fourier masses. This does not stabilize its pre-existing metric moduli.

The poles are local minima, not global minima. On the unit sphere,

$$\frac{\ell^2}{12}\mathcal{V}_\eta + 1 = 2(X^5)^2 + (X^6)^2 + \eta(X^6)^2 R_5^2 \geq 0 \quad (\eta \geq 0). \tag{17.8}$$

The global minimum $-12/\ell^2$ lies on $X^5 = X^6 = 0$, a vacuum S^3. No lifetime or transition rate for a pole background is calculated here.

18 D: the equatorial theory and a dynamical core

D is the equatorial restriction of the same parent action. It retains an independent one-time bulk and adds nonconstant order-parameter stress. Within its regular defect sector, the physical real slice is a field-defined core rather than a selected component of a prescribed reflection. The exact noncompact warped solution below is one solution family of the full D equations.

18.1 Fields and independent assumptions

Let $\widehat{M}$ be an eight-dimensional real manifold carrying a Lorentzian metric $\widehat{g}$ with one time direction. In a neighborhood of the proposed core use coordinates (x^μ, y^A), with $\mu = 0, 1, 2, 3$ and $A = 1, 2, 3, 4$. The normal coordinates have dimensions of length and are noncompact in the explicit solution.

In addition to the CR fields Q, introduce a real unit five-vector

$$N^a N^a = 1, \qquad a = 1, \ldots, 5. \tag{18.1}$$

Its target is the unit sphere S^4, with kinetic metric $6\mathrm{d}\Omega_4^2$. There are four independent scalar degrees of freedom in N. This target sphere is not the normal spacetime: it is the space of values of the defect field.

Choose a positive length ℓ and the potential

$$\mathcal{U}(N) = \frac{24(N^5)^2 - 12}{\ell^2}. \tag{18.2}$$

It is bounded on the sphere,

$$-\frac{12}{\ell^2} \le \mathcal{U} \le \frac{12}{\ell^2}, \tag{18.3}$$

and its vacuum manifold is the equator $N^5 = 0$, an S^3. The potential preserves rotations of the first four components and the symmetry $N^5 \mapsto -N^5$. The coefficients and ℓ are specified couplings. They will be justified by an inverse construction of the exact solution, not claimed to follow uniquely from the CR postulates.

18.2 Action and field equations

The bulk action is

$$S_D = \frac{M_8^6}{2} \int_{\widehat{M}} \mathrm{vol}_{\widehat{g}} \Big[\widehat{R} - \mathfrak{h}_{IJ}(Q) \partial_M Q^I \partial^M Q^J - 6 \partial_M N^a \partial^M N^a \\ - 2\mathcal{U}(N) + \lambda_D (N^a N^a - 1) \Big]. \tag{18.4}$$

The multiplier enforces the target constraint; its contribution to the stress vanishes after that constraint is imposed. The trace-reversed Einstein equation is

$$\widehat{R}_{MN} = \mathfrak{h}_{IJ} \partial_M Q^I \partial_N Q^J + 6 \partial_M N^a \partial_N N^a + \frac{\mathcal{U}}{3} \widehat{g}_{MN}. \tag{18.5}$$

The factor $1/3$ is $2/(D-2)$ at $D = 8$. The CR fields obey

$$\widehat{\Box} Q^I + \Gamma^I{}_{JK}(\mathfrak{h}) \widehat{g}^{MN} \partial_M Q^J \partial_N Q^K = 0. \tag{18.6}$$

Before eliminating the multiplier, variation of N^a gives

$$6 \widehat{\Box} N^a - \mathcal{U}_{,a} + \lambda_D N^a = 0, \qquad \lambda_D = 6(\partial N)^2 + N^a \mathcal{U}_{,a}. \tag{18.7}$$

The second identity follows by contraction with N^a, using $N^a\widehat{\Box}N^a = -(\partial N)^2$. Eliminating λ_D gives the projected sphere equation

$$\widehat{\Box}N^a + (\partial N)^2 N^a - \frac{8N^5}{\ell^2}\left(\delta^{a5} - N^5 N^a\right) = 0, \qquad (\partial N)^2 = \partial_M N^b \partial^M N^b. \tag{18.8}$$

The force is tangent to S^4; contraction with N^a gives the identity implied by the unit constraint.

The sigma kinetic metrics are positive on their real targets. For a real null vector k,

$$\widehat{R}_{MN}k^M k^N = \mathfrak{h}_{IJ}(k\cdot\partial Q^I)(k\cdot\partial Q^J) + 6\sum_a (k\cdot\partial N^a)^2 \geq 0. \tag{18.9}$$

Thus the negative value of the asymptotic potential does not violate the null energy condition. Local harmonic-gauge evolution has the usual one-time Einstein–sigma-model principal part. These observations are not a stability theorem for the defect: the coupled scalar and metric perturbations still require analysis.

18.3 The equator recovers the full defect theory

Set

$$X^6 = 0, \qquad X^a = N^a, \qquad N^a N^a = 1. \tag{18.10}$$

The equator is the fixed set of the isometry $X^6 \mapsto -X^6$ of the round target, and is totally geodesic. The potential has the same reflection symmetry. In particular,

$$\mathcal{V}_{\eta,6} = \frac{24\eta}{\ell^2} X^6 R_5^2, \tag{18.11}$$

so the discarded sixth equation vanishes identically on the equator.

Off the five-vector constraint, the restricted potential obeys

$$\mathcal{V}_\eta(N,0) = \mathcal{U}(N) - \frac{12}{\ell^2}(N^a N^a - 1). \tag{18.12}$$

Consequently the correct identification of multipliers is

$$\lambda_D = \Lambda + \frac{24}{\ell^2}. \tag{18.13}$$

With this shift, the restricted action is exactly (18.4), including its off-constraint multiplier convention.

Proposition 18.1 (Equatorial reduction)**.** *Every solution of the original D equations lifts to a solution of S_* by (18.10) and (18.13), at the same fixed ℓ, M_8, η. This holds for the complete D theory, not only its displayed spherical defect solution.*

Proof. The unit constraint and kinetic terms agree. On the constraint surface the potential, Einstein equations, and CR equations agree. For $A = 1,\ldots,4$, the parent potential derivative is $-24N^A/\ell^2$, while the original derivative is zero. For component five these derivatives are $24N^5/\ell^2$ and $48N^5/\ell^2$. In both cases the shift (18.13) makes the unprojected equations equal. The sixth equation vanishes by (18.11). These exhaust the independent equations. □

Initial data lying in either restriction, with zero normal target velocity, remain there under local uniqueness of the corresponding regular hyperbolic evolution. This invariance does not assert that generic initial data are attracted to either sector. The full common theory has $25+n$ local physical configuration degrees of freedom: 20 gravitational, n CR scalars, and five sphere coordinates. B and D retain $20+n$ and $24+n$, respectively. They are consistent subsectors, not invertible field redefinitions of the full theory.

19 An explicit smooth codimension-four solution

Define

$$r^2 = \delta_{AB} y^A y^B, \qquad s = r/\ell, \qquad \Omega(r) = \frac{1}{\sqrt{1+s^2}}, \qquad A(r) = \log \Omega(r). \tag{19.1}$$

The proposed configuration is

$$\mathrm{d}\widehat{s}^2 = \Omega^2(r) \left[g_{\mu\nu}(x) \mathrm{d}x^\mu \mathrm{d}x^\nu + \delta_{AB} \mathrm{d}y^A \mathrm{d}y^B \right], \tag{19.2}$$

$$N^A = \frac{y^A}{\sqrt{\ell^2 + r^2}}, \qquad N^5 = \frac{\ell}{\sqrt{\ell^2 + r^2}}, \qquad Q^I = Q^I(x). \tag{19.3}$$

At $r = 0$, the metric is smooth, $\Omega = 1$, and $N = (0, 0, 0, 0, 1)$. In particular,

$$\partial_B N^A|_{r=0} = \frac{\delta^A_B}{\ell}. \tag{19.4}$$

The first four components have a regular common zero. Their zero set is therefore a four-manifold, and the induced metric is exactly g.

19.1 Einstein equations in components

Let Δ and inner products of normal derivatives refer to the flat δ_{AB} metric. The conformal curvature formula in eight dimensions gives

$$\widehat{R}_{\mu\nu} = R_{\mu\nu}[g] - g_{\mu\nu}(\Delta A + 6|\nabla A|^2), \tag{19.5}$$

$$\widehat{R}_{\mu A} = 0, \tag{19.6}$$

$$\widehat{R}_{AB} = -6(\partial_A \partial_B A - \partial_A A \partial_B A) - \delta_{AB}(\Delta A + 6|\nabla A|^2). \tag{19.7}$$

The defect derivatives obey the exact identities

$$\partial_A N^a \partial_B N^a = \frac{\delta_{AB}}{\ell^2 + r^2} - \frac{y_A y_B}{(\ell^2 + r^2)^2}, \tag{19.8}$$

$$-6(\partial_A \partial_B A - \partial_A A \partial_B A) = 6 \partial_A N^a \partial_B N^a, \tag{19.9}$$

$$-(\Delta A + 6|\nabla A|^2) = \frac{\mathcal{U}}{3}\Omega^2 = \frac{4(\ell^2 - r^2)}{(\ell^2 + r^2)^2}. \tag{19.10}$$

These equations already check the internal and mixed Einstein components. The remaining external equation is simply

$$R_{\mu\nu}[g] = \mathfrak{h}_{IJ}(Q) \partial_\mu Q^I \partial_\nu Q^J. \tag{19.11}$$

Unlike the earlier flat-torus model, a nonzero potential is now balanced by normal gradients and curvature. It has not been added while keeping the old internal geometry fixed.

19.2 The scalar equations

For an internally constant CR field,

$$\widehat{\Box} Q^I = \Omega^{-2} \Box_g Q^I. \tag{19.12}$$

The target-connection term has the same overall factor. Equation (18.6) therefore reduces to the wave-map equation of (9.8).

For a defect component depending only on y,

$$\widehat{\Box} N^a = \Omega^{-2} \left(\Delta N^a + 6 \nabla A \cdot \nabla N^a \right), \qquad (\partial N)^2 = \frac{4 + 3s^2}{\ell^2 (1 + s^2)}. \tag{19.13}$$

Substitution in (18.8) gives zero for all five components. For example,

$$\widehat{\Box}N^5 = \frac{-4+5s^2}{\ell^2(1+s^2)^{3/2}}. \tag{19.14}$$

Adding $(\partial N)^2 N^5$ gives $8s^2/[\ell^2(1+s^2)^{3/2}]$, which is precisely the projected potential force. The four remaining components obey the same projected equation. The multiplier completing the unprojected system is

$$\lambda_D = \frac{18(4+s^2)}{\ell^2(1+s^2)}. \tag{19.15}$$

The parent multiplier is $\Lambda = \lambda_D - 24/\ell^2 = (-6+54\Omega^2)/\ell^2$. The verification scripts check the projected equations and, at two nonflat seed points, all five unprojected D equations with λ_D; the common-parent suite also checks all six equations with Λ.

20 Exact recovery of the CR equations

Theorem 20.1 (Warped CR embedding)**.** *Let (g, Q) solve the potential-free four-dimensional action (9.8). Then (19.2)–(19.3) solve all equations of the bulk action (18.4) at every finite normal coordinate on the physical coordinate patch. Conversely, a solution within this ansatz restricts to the same CR equations. No small-radius, small-curvature, or low-energy condition is used.*

Proof. Equations (19.9) and (19.10) satisfy the normal Einstein equations exactly; the mixed equations vanish on both sides. The external components become (19.11). The CR wave-map equations acquire only the nonzero common factor Ω^{-2}, and the defect equations are satisfied independently of the slice fields. These are all independent bulk equations, including the unit constraint and its multiplier equation. The same calculations prove the converse within the ansatz. □

The proof concerns a solution sector, rather than generic bulk data. Smooth slice solutions suffice for this real bulk embedding; the later holomorphic reconstruction additionally requires real analyticity. No assertion is made that the asymptotic end of every lifted solution is regular. Section 25 gives an explicit failure of that stronger statement. These qualifications are necessary even though the finite-coordinate field equations are exact.

20.1 Finite four-dimensional normalization

The coefficient of $R[g]$ in the bulk action is

$$I_6 = \int_{\mathbb{R}^4} \mathrm{d}^4y\, \Omega^6 = 2\pi^2 \int_0^\infty \frac{r^3 \mathrm{d}r}{(1+r^2/\ell^2)^3} = \frac{\pi^2\ell^4}{2}, \tag{20.1}$$

$$M_{\mathrm{P}}^2 = M_8^6 I_6 = \frac{\pi^2}{2} M_8^6 \ell^4. \tag{20.2}$$

The same integral multiplies the CR target kinetic term. Consequently the canonical normalization (9.6), including its geometric scalar coefficient, agrees with the preceding CR model.

The internal proper volume is infinite: $\int \mathrm{d}^4y\, \Omega^4$ diverges logarithmically. The finite result in (20.1) comes from warping, not a compact normal manifold. The length ℓ is a coupling in the specified potential; its value has not been calculated from another CR principle.

20.2 The background term cancels after integration

Substituting a solution ansatz into an action can lose field equations [11]; that is why Theorem 20.1 was proved first. Here the action calculation provides a further useful check.

For arbitrary off-shell slice fields, substituting the fixed defect profile separates the bulk Lagrangian into $\Omega^{-2}[R[g] - \mathfrak{h}_{IJ}\partial Q^I \partial Q^J]$ and a background term. On this profile the latter is $2\mathcal{U}/3$. With $u = r^2/\ell^2$, its integrated coefficient is

$$\int \mathrm{d}^4y\, \Omega^8 \frac{2\mathcal{U}}{3} = 8\pi^2\ell^2 \int_0^\infty \frac{u(1-u)}{(1+u)^5}\mathrm{d}u = 0. \tag{20.3}$$

Thus the restricted action is exactly (9.8), with no residual four-dimensional cosmological term. The ordinary radial gravitational boundary term vanishes at both ends of the regulated integral for this profile.

This cancellation is part of the tuned model. An arbitrary additive shift in $\mathcal{U}$ changes the field equations and need not preserve the solution or (20.3). It is not a solution of the cosmological constant selection problem.

21 What makes the core a physical real slice

21.1 A covariant definition

In the regular north-core sector define

$$M_N = \{p \in \widehat{M} : N^1 = N^2 = N^3 = N^4 = 0,\ N^5 > 0\}. \tag{21.1}$$

This definition uses scalar fields, so it is independent of a chosen coordinate label. If their four gradients are independent, the implicit-function theorem gives a codimension-four submanifold. Its embedding varies when the defect fields vary. A general solution must additionally have a Lorentzian induced metric and spacelike normal directions; the explicit solution satisfies these conditions.

The action determines the evolution of N, but does not select an absolute position in a diffeomorphism-invariant spacetime. Nor does it force every solution into the same defect sector. The achievement is a material, dynamical definition of the slice within a specified solution sector.

21.2 The role and limit of the winding number

At normal infinity the explicit profile tends to

$$N^5 \longrightarrow 0, \qquad N^A \longrightarrow y^A/r. \tag{21.2}$$

The boundary map from the normal S^3 to the vacuum S^3 has degree one. Holding this asymptotic sector fixed, the first four fields cannot remain nonzero everywhere in the normal ball: otherwise their normalized vector would extend a degree-one boundary map to the ball. At a zero of the four components, the unit constraint requires $N^5 = \pm 1$.

This topological argument guarantees at least one zero of the first four components under the stated boundary conditions. It does not establish a regular four-dimensional core. For example, the smooth sphere-valued test field

$$\zeta^A = \frac{r^2 y^A}{\ell^3}, \qquad \widetilde{N} = \frac{(\zeta^1, \ldots, \zeta^4, 1)}{\sqrt{1 + r^6/\ell^6}} \tag{21.3}$$

has the same degree-one asymptotic map, but $\partial_B \widetilde{N}^A = 0$ at its central zero. This is a kinematic counterexample, not an asserted solution of the defect equations. Rank, Lorentzian induced geometry, and maximal reality must be checked separately; they hold for the displayed exact profile. Degree alone also proves neither uniqueness nor stability, chooses neither north nor south, and does not force evolution into this sector from arbitrary data. The asymptotic vacuum map is an added boundary condition, not a consequence of finite warped integrals.

21.3 The local real structure can follow the core

Retain an integrable complex structure J on the underlying real eight-manifold. Near the explicit solution choose local holomorphic coordinates

$$z^\mu = x^\mu + \mathrm{i}y^{\mu+1}, \qquad \mu = 0, \dots, 3. \tag{21.4}$$

The core is maximally real. Locally, a sufficiently C^1-small deformation remains maximally real, although a general large deformation need not. If the fields and core are real analytic, an adapted holomorphic chart straightens the core to a real coordinate locus. Conjugation in that chart defines an antiholomorphic involution σ_N whose fixed locus is the core [6].

The germ of this reflection is unique: the composition of two such antiholomorphic maps would be holomorphic and fix an open maximally real slice pointwise, hence would be the identity. In this local analytic sense, the core determines the real structure once J is supplied. The construction still has no field equation selecting J, and it does not prove a global reflection exists.

21.4 Total geodesy in the symmetric sector

For the exact profile, simultaneous normal inversion $y \mapsto -y$ is an isometry of $\widehat{g}$. It acts on the defect fields as $N^A \mapsto -N^A$, $N^5 \mapsto N^5$, an internal symmetry of the action. The CR fields are even. At the core,

$$\partial_A \Omega = 0, \qquad \widehat{\Gamma}^A{}_{\mu\nu} = 0. \tag{21.5}$$

The core is totally geodesic. A free bulk geodesic initially tangent to it remains there by uniqueness. Initial data respecting the combined spacetime and target reflection retain that symmetry under local evolution in a compatible gauge.

For a general fluctuating defect, its zero set can bend. Field-defined does not automatically mean totally geodesic. The free-particle preservation statement applies to the symmetric sector just described; arbitrary defect solutions require their extrinsic geometry and matter forces to be checked.

22 Tensor and response localization

22.1 The normal operator

Take the flat seed $g = \eta$ and constant CR fields $Q = Q_0$. To check decoupling explicitly, set

$$\begin{gathered} \delta\widehat{g}_{\mu\nu} = \Omega^2 h_{\mu\nu}, \qquad \delta\widehat{g}_{\mu A} = \delta\widehat{g}_{AB} = 0, \qquad \delta N = \delta Q = 0, \\ \partial^\mu h_{\mu\nu} = 0, \qquad \eta^{\mu\nu} h_{\mu\nu} = 0. \end{gathered} \tag{22.1}$$

Indices on h are raised with η. With $\mathcal{B} = \Delta A + 6|\nabla A|^2$, the linearized external Ricci tensor is

$$\delta\widehat{R}_{\mu\nu} = -\frac{1}{2}\left(\Box_4 + \Delta + 6\partial_A A\, \partial_A\right) h_{\mu\nu} - \mathcal{B} h_{\mu\nu}. \tag{22.2}$$

The external source variation is $(\mathcal{U}/3)\Omega^2 h_{\mu\nu} = -\mathcal{B} h_{\mu\nu}$, by the background equation. These algebraic terms cancel. The mixed and internal linear Einstein equations vanish, and the

sphere equations acquire no first-order source: the external trace and divergence in (22.1) remove their possible contributions. The remaining tensor equation is therefore

$$(\Box_4 + \Delta + 6\partial_A A\,\partial_A)\,h_{\mu\nu} = 0. \tag{22.3}$$

Its derivative weight in the quadratic action is Ω^6, in agreement with the thick-brane tensor analysis [12]. Writing $h_{\mu\nu} = \epsilon_{\mu\nu}(x)\psi(y)$, with $\Box_4\epsilon_{\mu\nu} = m_\perp^2\epsilon_{\mu\nu}$ for the stated signature, gives

$$\mathcal{L}_\perp\psi = m_\perp^2\psi, \qquad \mathcal{L}_\perp = -\Omega^{-6}\partial_A(\Omega^6\partial_A), \tag{22.4}$$

with Hilbert norm

$$\|\psi\|^2 = \int_{\mathbb{R}^4} \mathrm{d}^4y\,\Omega^6|\psi|^2. \tag{22.5}$$

Linear CR response perturbations about constant Q_0, after a constant positive target normalization, obey the same normal equation. This statement does not include the coupled defect and scalar-metric perturbations.

Start with the nonnegative form on smooth compactly supported profiles. Its closure defines the Friedrichs operator, and for profiles in the operator domain,

$$\langle\psi, \mathcal{L}_\perp\psi\rangle = \int \mathrm{d}^4y\,\Omega^6|\nabla\psi|^2 \geq 0. \tag{22.6}$$

The constant profile is a normalizable zero mode. With unit norm it is

$$\psi_0 = \frac{\sqrt{2}}{\pi\ell^2}. \tag{22.7}$$

The domain statement can be checked explicitly. For the normalized constant profile, the weighted norm outside a radius R is

$$T(R) = \int_{|y|>R} \mathrm{d}^4y\,\Omega^6|\psi_0|^2 = \frac{1+2(R/\ell)^2}{[1+(R/\ell)^2]^2}. \tag{22.8}$$

Choose a smooth cutoff equal to one for $r \leq R$, zero for $r \geq 2R$, and with gradient bounded by c/R. The norm error tends to zero, and its form energy is at most $c^2T(R)/R^2 \to 0$. Thus ψ_0 belongs to the closed form domain. Its bilinear form pairing with every test profile is zero, so it also belongs to the associated operator domain and is a zero eigenfunction. Conversely, zero form energy implies a constant profile on connected $\mathbb{R}^4$. The zero normal profile is unique within this block.

22.2 Schrödinger form and a gapless continuum

The unitary change $\widetilde{\psi} = \Omega^3\psi$ gives the flat-measure operator

$$\mathsf{H}_\perp = -\Delta + V_\perp(r),$$

$$V_\perp(r) = 3\Delta A + 9|\nabla A|^2 = \frac{3(r^2-4\ell^2)}{(\ell^2+r^2)^2}, \tag{22.9}$$

$$\mathsf{H}_\perp = \sum_A(-\partial_A - 3\partial_A A)(\partial_A - 3\partial_A A). \tag{22.10}$$

The zero wavefunction is proportional to Ω^3. Although $V_\perp(0) = -12/\ell^2$, the factorization and the weighted quadratic form exclude negative $m_\perp^2$ in this block. A negative region of a Schrödinger potential is not itself a tachyon test.

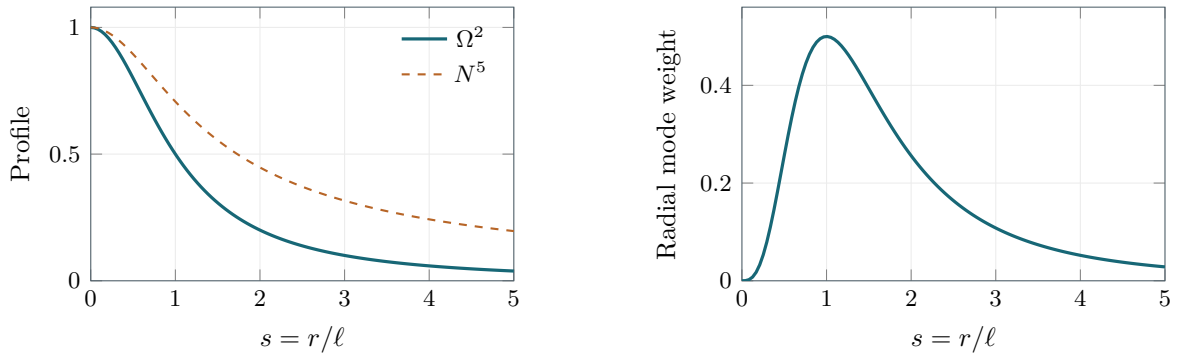


Figure 1: Exact profiles of the defect and localized zero mode. The right panel gives the normalized radial weight in the classical mode inner product: $p(s) = 4s^3/(1+s^2)^3$, with $\int_0^\infty p(s)\mathrm{d}s = 1$. It is not an independently established quantum probability. Its maximum at $s = 1$ is a shell-volume effect; Ω^3 itself is largest at the core. The displayed range omits the remaining tail.

Here $V_\perp$ is smooth and bounded,

$$-\frac{12}{\ell^2} \le V_\perp \le \frac{3}{20\ell^2}, \qquad V_\perp(r) = \frac{3}{r^2} + O(r^{-4}). \tag{22.11}$$

Consequently $\mathsf{H}_\perp$ is the self-adjoint bounded-potential perturbation of $-\Delta$ with domain $H^2(\mathbb{R}^4)$. It is essentially self-adjoint on C_c^∞ and agrees with the transformed Friedrichs operator. Multiplication by $V_\perp$ is relatively compact: a compactly supported truncation composed with the free resolvent is compact by local Sobolev compactness, and the discarded tail tends to zero in operator norm. Weyl's theorem therefore gives $\sigma_{\mathrm{ess}}(\mathsf{H}_\perp) = [0, \infty)$ [14]. Together with positivity, this places the normalizable zero state at the continuum threshold. There is no positive mass gap.

The exact embedding theorem does not rely on such a gap. It specifies a preserved classical sector. A generic bulk perturbation need not stay in that sector, and finite normalization alone does not calculate the coupling of continuum modes to physical sources or establish an acceptable measured gravitational force law.

22.3 What the mode result does not prove

The unrestricted bulk theory has 20 gravitational polarizations, n CR scalars, and four independent sphere fields: $24 + n$ local configuration degrees of freedom before selecting a background sector. The preceding calculation studies its external tensor block and the linear CR response block around constant Q_0. It does not classify localized bending modes, coupled scalar modes, or vector modes.

In particular, a bounded potential and positive target metric do not prove that the full defect is stable. The north pole is a maximum of the potential along the sphere; its stability as part of an inhomogeneous topological configuration depends on gradient and gravitational terms. A complete constrained fluctuation operator is still needed. No identification with chiral matter or the observed gauge interactions follows from the tensor zero mode.

23 The additional parent scalar about D

23.1 The decoupled odd perturbation

Write local latitude coordinates

$$X = (\cos\alpha\, N, \sin\alpha), \qquad N \cdot N = 1. \tag{23.1}$$

Then

$$\mathrm{d}X \cdot \mathrm{d}X = \mathrm{d}\alpha^2 + \cos^2\alpha\, \mathrm{d}N \cdot \mathrm{d}N, \tag{23.2}$$

$$\mathcal{V}_\eta = \cos^2\alpha\, \mathcal{U}(N) + \frac{12\eta}{\ell^2}\sin^2\alpha\cos^2\alpha. \tag{23.3}$$

The old defect is $\alpha = 0$. Reflection makes α odd and all background metric, CR, and equatorial variables even. There are no bilinear couplings between this odd perturbation and those even perturbations. Moreover, $\delta\alpha$ is gauge invariant under linearized spacetime diffeomorphisms because its background is constant.

Put $K_N = (\partial N)^2$. At quadratic order the odd action is

$$S^{(2)}_{\mathrm{odd}} = -3M_8^6 \int \mathrm{vol}_{\widehat{g}} \left[(\partial\alpha)^2 + m^2_{\mathrm{eff}}(y)\alpha^2\right], \quad m^2_{\mathrm{eff}} = \frac{4\eta}{\ell^2} - K_N - \frac{\mathcal{U}}{3}. \tag{23.4}$$

On the flat defect, $g = \eta_{\mu\nu}$, $Q = Q_0$,

$$K_N = \frac{3+\Omega^2}{\ell^2}, \qquad m^2_{\mathrm{eff}} = \frac{4\eta + 1 - 9\Omega^2}{\ell^2}. \tag{23.5}$$

The same equation follows independently by linearizing the sixth unprojected equation (10.7) and using $\Lambda = \lambda_D - 24/\ell^2$ and (19.15).

With $\alpha = \epsilon(x)\psi(y)$ and $\Box_4\epsilon = m_\perp^2\epsilon$, the weighted normal operator is

$$L_\eta = -\Omega^{-6}\partial_A(\Omega^6\partial_A) + \Omega^2 m^2_{\mathrm{eff}}, \qquad \|\psi\|^2 = \int \Omega^6|\psi|^2\mathrm{d}^4y. \tag{23.6}$$

The unitary change $f = \Omega^3\psi$ gives

$$H_\eta = -\Delta + W_\eta(r), \tag{23.7}$$

$$W_\eta(r) = \frac{4[(\eta+1)r^2 + (\eta-5)\ell^2]}{(\ell^2+r^2)^2}. \tag{23.8}$$

23.2 An exact threshold, not a pointwise-potential guess

For each finite real η, W_η is real, smooth, bounded, and tends to zero. The self-adjoint realization on $H^2(\mathbb{R}^4)$ is the bounded-potential perturbation of $-\Delta$. Its essential spectrum is $[0,\infty)$: multiplication by a compactly supported truncation of W_η composed with the free resolvent is compact, and the tail tends to zero in operator norm. These are the same domain and relative-compactness arguments used in the integrated audit; see also [14].

A useful exact identity is

$$H_\eta = \sum_{A=1}^{4}(-\partial_A - 4\partial_A\log\Omega)(\partial_A - 4\partial_A\log\Omega) + \frac{4(\eta-1)}{\ell^2+r^2}. \tag{23.9}$$

Theorem 23.1 (Sharp threshold for the new odd block)**.** *On the original flat-defect patch with the stated self-adjoint domain:*

1. *If $\eta > 1$, the odd block has no negative eigenvalues and no normalizable zero eigenfunction. Its spectrum still begins at zero; there is no positive gap.*
2. *If $\eta = 1$, it is nonnegative and has a normalizable zero state $f_0 = \Omega^4$, or $\psi_0 = \Omega$ in the weighted representation.*
3. *If $\eta < 1$, it has at least one negative eigenvalue.*

Proof. On compactly supported smooth functions, (23.9) gives a sum of square norms plus $4(\eta-1)\int |f|^2/(\ell^2+r^2)$. The statement extends to the closed form domain. For $\eta > 1$, vanishing of the form would force $f = 0$, excluding a zero eigenfunction; nonnegativity excludes negative spectrum.

At $\eta = 1$, each first-order factor annihilates $f_0 = \Omega^4$. This smooth function lies in $H^2(\mathbb{R}^4)$, since it and its first two derivatives decay sufficiently fast, and

$$\|f_0\|^2 = \int_{\mathbb{R}^4} \Omega^8 \mathrm{d}^4 y = \frac{\pi^2 \ell^4}{6}. \tag{23.10}$$

Zero form energy also implies f is proportional to Ω^4 on the connected normal space, so this zero profile is unique in the odd scalar block.

For $\eta < 1$, the same admissible trial function gives

$$\frac{\langle f_0, H_\eta f_0\rangle}{\|f_0\|^2} = \frac{2(\eta-1)}{\ell^2} < 0. \tag{23.11}$$

Since the essential spectrum starts at zero, the variational principle supplies a discrete negative eigenvalue. The essential spectrum remains $[0, \infty)$ in all cases. □

The coincident threshold $\eta = 1$ for the four pole masses and the defect's new odd mode is a useful selection criterion for this family. The spectral result permits the strictly wider range $\eta > 1$, whereas demanding $m^2_{\text{eff}} \geq 0$ pointwise would require $\eta \geq 2$. For example, at $\eta = 3/2$ the core value is $m^2_{\text{eff}}(0) = -2/\ell^2$, yet the factorization proves that the normal operator is nonnegative and has no normalizable zero state. A negative local mass term or negative portion of W_η is not by itself a negative spectral eigenvalue.

The strict choice $\eta > 1$ avoids the marginal mode. Its absence does not establish nonlinear stability. In particular, the old coupled defect-metric scalar and vector blocks remain to be analyzed. The old tensor block is unchanged, and the new odd block decouples only at the linearized level about the equatorial background.

23.3 The threshold does not settle nonlinear stability

At $\eta = 1$, a rigid rotation of the defect in the $(5, 6)$ plane illustrates why the zero quadratic mode is not a nonlinear stability theorem. Keeping the metric fixed as a trial configuration, put $X^5 = \cos\epsilon\,\Omega$, $X^6 = \sin\epsilon\,\Omega$, with X^A unchanged. The kinetic norm is unchanged, while

$$\Delta\mathcal{V} = \frac{12}{\ell^2}\left[(\eta-1)\sin^2\epsilon\,\Omega^2 - \eta\sin^4\epsilon\,\Omega^4\right]. \tag{23.12}$$

The quartic term is negative at the threshold. This trial family does not impose the coupled gravitational constraints and is not a proof of nonlinear instability. It is a reason to keep the proven claim at the level of the quadratic block and to use the strict range $\eta > 1$ for the candidate.

24 Massive probes: preservation, repulsion, and local binding

24.1 A totally geodesic core can still repel

For a free massive probe, perturb a timelike geodesic of the core by a small normal displacement y^A. At the core $g_{\mu\nu}u^\mu u^\nu = -1$, and

$$\widehat{\Gamma}^A{}_{\mu\nu} = \frac{y^A}{\ell^2 + r^2} g_{\mu\nu}. \tag{24.1}$$

To linear order in normal displacement, with core proper time τ,

$$\frac{\mathrm{d}^2 y^A}{\mathrm{d}\tau^2} = \frac{y^A}{\ell^2}. \tag{24.2}$$

The exactly tangent orbit stays in the core, but a displaced massive free probe is repelled. Gravity by itself has not supplied stable material observers on the slice.

This is compatible with tensor localization. The tensor equation is a wave eigenvalue problem with a warped norm; massive free-particle motion is a different dynamical test. One result cannot be substituted for the other.

24.2 An explicit, regular probe coupling

Consider the real worldline action

$$S_p = \int \left[P_M \dot{Z}^M - \frac{e}{2}\left(\widehat{g}^{MN} P_M P_N + m_{\mathrm{eff}}^2(N) \right) \right] \mathrm{d}\tau, \tag{24.3}$$

with the smooth positive mass function

$$m_{\mathrm{eff}}^2(N) = m^2 \left[1 + \beta(1 - (N^5)^2) \right], \qquad m > 0, \quad \beta > 0. \tag{24.4}$$

On the core $m_{\mathrm{eff}} = m$; its tangential derivatives and first normal derivatives vanish. The core shell and its real free flow therefore remain (1.3). The complex representative remains compatible with P2.

The linear transverse equation becomes

$$\frac{\mathrm{d}^2 y^A}{\mathrm{d}\tau^2} + \frac{\beta - 1}{\ell^2} y^A = 0. \tag{24.5}$$

For $\beta > 1$, the coupling overcomes gravitational repulsion and supplies a restoring force, with frequency $\sqrt{\beta - 1}/\ell$. This result holds for perturbations of a core timelike trajectory within the displayed warped solution. A universal β would make this leading normal acceleration independent of the rest mass, but universality is an additional assumption.

24.3 The binding is local and has a finite escape barrier

For a Minkowski slice, conserved energy E, tangential momentum $\mathbf{p}$, and zero normal angular momentum reduce the shell to

$$E^2 - |\mathbf{p}|^2 = p_r^2 + W_\beta(r), \qquad W_\beta(r) = m^2 \frac{1 + (1+\beta)s^2}{(1+s^2)^2}. \tag{24.6}$$

For $\beta > 1$, the core is a local minimum of W_β. The barrier is at

$$s_b^2 = \frac{\beta - 1}{\beta + 1}, \qquad W_\beta(s_b) = m^2 \frac{(\beta+1)^2}{4\beta}, \tag{24.7}$$

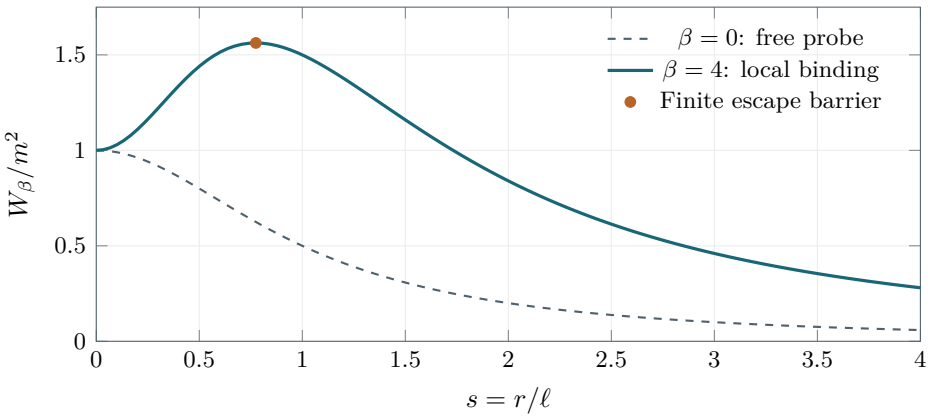


Figure 2: The same defect geometry gives transverse repulsion for free massive probes and local binding for the explicit mass coupling. For $\beta = 4$, the barrier has height $25m^2/16$ at $r = \ell\sqrt{3/5}$. Both potentials tend to zero at infinity. The figure is an exact plot of (24.6), not a simulation or a fitted phenomenological curve.

while $W_\beta \to 0$ as $r \to \infty$. The barrier rises above the core value by $W_\beta(s_b) - m^2 = m^2(\beta - 1)^2/(4\beta)$. Low-energy trajectories in the central allowed region can be trapped classically below the barrier. Sufficiently energetic trajectories can escape. The result is local binding, not absolute confinement. A quantum tunnelling rate has not been calculated.

The borderline $\beta = 1$ deserves a separate check. Its linear restoring frequency vanishes, but with $u = s^2$,

$$\frac{W_1}{m^2} = \frac{1+2u}{(1+u)^2} = 1 - \frac{u^2}{(1+u)^2} = 1 - s^4 + O(s^6). \tag{24.8}$$

The core is a quartically repulsive maximum in this case, not a nonlinearly neutral bound state. The local-binding claim requires the strict inequality $\beta > 1$.

The coupling is a test-particle interaction. It does not establish a fully backreacting localized matter sector or extend Theorem 20.1 to arbitrary localized sources. For $m = 0$, the mass function in (24.4) vanishes identically and supplies no such binding. Gauge fields, fermions, and physical detectors require their own field actions. The parameter β is not fixed by P1, P2, or the defect equations.

25 Global geometry and a limit of the exact uplift

25.1 A smooth core and bounded flat-seed invariants

For $g = \eta$ and constant CR fields, direct curvature calculation gives

$$\widehat{R} = \frac{14(4 - s^2)}{\ell^2(1 + s^2)}, \tag{25.1}$$

$$\widehat{R}_{MN}\widehat{R}^{MN} = \frac{464 - 88s^2 + 92s^4}{\ell^4(1 + s^2)^2}, \tag{25.2}$$

$$\widehat{R}_{MNPQ}\widehat{R}^{MNPQ} = \frac{160 + 16s^2 + 52s^4}{\ell^4(1 + s^2)^2}. \tag{25.3}$$

They are finite at the core and at large r. In particular, $\widehat{R} \to -14/\ell^2$, while the Riemann square tends to $52/\ell^4$.

With $r = \ell \sinh \rho$, the flat-seed metric is

$$\mathrm{d}\widehat{s}^2 = \operatorname{sech}^2 \rho \, \mathrm{d}s^2_{1,3} + \ell^2 \mathrm{d}\rho^2 + \ell^2 \tanh^2 \rho \, \mathrm{d}\Omega_3^2. \tag{25.4}$$

The normal proper radial distance diverges as $\rho \to \infty$, while the geometry approaches an anti-de Sitter five-dimensional throat times a three-sphere, both with curvature radius ℓ. This explains the limiting scalar curvature $-20/\ell^2 + 6/\ell^2$.

25.2 Bounded invariants are not geodesic completeness

A radial null geodesic with conserved positive $E = \Omega^2 \dot{t}$ obeys $\dot{r} = E/\Omega^2$. Its remaining affine length is

$$\lambda_\infty - \lambda(r) = \frac{\ell}{E} \left[\frac{\pi}{2} - \arctan(r/\ell) \right]. \tag{25.5}$$

It reaches the asymptotic end in finite affine parameter even though the coordinate time diverges. The displayed patch is therefore null-geodesically incomplete. Bounded polynomial invariants alone would not decide whether this is an extendible patch horizon or a curvature singularity. For the flat seed, the following construction resolves the local question explicitly. Maximal global completion remains a separate problem.

The normal-mode analysis in Section 22 uses the specified patch and its natural spatial operator domain. Extending the background does not automatically extend its separated perturbation modes or preserve their global norm. Those questions must be checked on the chosen larger spacetime.

25.3 A regular local extension of the flat end

For the flat seed with constant Q, set $u = \Omega = N^5$ on the original patch, where $0 < u \leq 1$. Away from the core,

$$r = \frac{\ell \sqrt{1-u^2}}{u}, \qquad \mathrm{d}\widehat{s}^2 = u^2 \eta_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu + \frac{\ell^2 \mathrm{d}u^2}{u^2(1-u^2)} + \ell^2 (1-u^2) \mathrm{d}\Omega_3^2. \tag{25.6}$$

The apparent singularity at $u = 0$ can be separated into a familiar coordinate singularity and a regular correction. Let $\bar{g}$ be the metric of AdS_5 with radius ℓ. On a Poincaré patch,

$$\bar{g} = u^2 \eta_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu + \frac{\ell^2 \mathrm{d}u^2}{u^2}. \tag{25.7}$$

Use its usual realization as a quadric in $\mathbb{R}^{2,4}$ [15]:

$$-X_{-1}^2 - X_0^2 + \sum_{i=1}^{3} X_i^2 + X_4^2 = -\ell^2, \qquad u = \frac{X_{-1} + X_4}{\ell}. \tag{25.8}$$

On the overlap, the coordinate map is

$$X_\mu = u x^\mu, \qquad X_{-1} + X_4 = \ell u, \qquad X_{-1} - X_4 = \frac{\ell}{u} + \frac{u}{\ell} \eta_{\mu\nu} x^\mu x^\nu. \tag{25.9}$$

Its induced metric is (25.7). The function u is regular across the Poincaré horizon; only the (u, x^μ) chart degenerates there. Locally on this quadric, or its universal cover, define

$$\widehat{g}_{\mathrm{ext}} = \bar{g} + \frac{\ell^2}{1-u^2} \mathrm{d}u^2 + \ell^2 (1-u^2) \mathrm{d}\Omega_3^2, \tag{25.10}$$

$$N = (\sqrt{1-u^2}\, \mathbf{n}, u), \qquad \mathbf{n} \in S^3, \qquad Q = Q_0. \tag{25.11}$$

These expressions are real analytic for $|u| < 1$, including $u = 0$, in regular local charts of the quadric and sphere. The remaining multiplier is regular as well:

$$\lambda_D = \frac{18(1+3u^2)}{\ell^2}, \qquad (\partial N)^2 = \frac{3+u^2}{\ell^2}. \tag{25.12}$$

Proposition 25.1 (Local extension of the flat defect)**.** *Equations* (25.10)–(25.11) *define a nondegenerate Lorentzian solution of the real defect equations on a neighborhood of the flat patch horizon. They agree with the original solution for* $0 < u < 1$ *and extend it through* $u = 0$.

Proof. On the quadric,

$$\bar{g}^{-1}(\mathrm{d}u, \mathrm{d}u) = \frac{u^2}{\ell^2}. \tag{25.13}$$

The rank-one update of the five-metric has determinant ratio

$$1 + \frac{\ell^2}{1-u^2}\bar{g}^{-1}(\mathrm{d}u, \mathrm{d}u) = \frac{1}{1-u^2} > 0. \tag{25.14}$$

Interpolating from $\bar{g}$ to the updated metric never crosses a zero determinant, so its one-time signature is retained. The sphere factor is positive. The full metric is therefore Lorentzian and nondegenerate across $u = 0$. Substitution of (25.7) gives (25.6); the fields and potential agree with the defect solution on the open set $u > 0$. All field-equation residuals are real analytic in the regular chart. They vanish on that open set, hence throughout a connected neighborhood across the horizon. This proves the extension without inferring regularity from scalar invariants alone. □

A regular coordinate chart makes the absence of a hidden determinant zero explicit. Put $X_{-1} - X_4 = v$, $X_i = Y_i$, and

$$S = \ell^2 + |Y|^2 - \ell u v > 0, \qquad X_0 = \sqrt{S}. \tag{25.15}$$

Then

$$\bar{g} = -\ell\,\mathrm{d}u\,\mathrm{d}v - \frac{(\mathrm{d}S)^2}{4S} + \sum_i \mathrm{d}Y_i^2, \qquad \det\left(\bar{g} + \frac{\ell^2 \mathrm{d}u^2}{1-u^2}\right) = -\frac{\ell^4}{4S(1-u^2)}. \tag{25.16}$$

At $u = 0$, the mixed metric component is $\bar{g}_{uv} = -\ell/2$. The audit script computes all eight-dimensional Einstein components and all five unprojected sphere equations directly in this regular chart, at two horizon points and a point with $u < 0$.

For the outgoing flat-seed ray with $x^i = 0$, fixed $\mathbf{n}$, and $t = r + t_*$, its regular coordinates are

$$\begin{aligned} &Y_i = 0, \qquad v = u\left(\ell - \frac{t_*^2}{\ell}\right) - 2t_*\sqrt{1-u^2}, \\ &X_0 = \ell\sqrt{1-u^2} + u t_*, \qquad \Delta\lambda = \frac{\ell}{E}\arcsin u, \qquad \frac{\mathrm{d}u}{\mathrm{d}\lambda} = -\frac{E}{\ell}\sqrt{1-u^2}. \end{aligned} \tag{25.17}$$

The endpoint $(u, v, Y) = (0, -2t_*, 0)$ and its tangent are finite. The geodesic therefore continues locally in the extended metric. Its Ricci contraction is $6E^2/\ell^2$, finite at the crossing.

This result concerns the real flat background and its defect fields. It proves neither a maximal geodesically complete spacetime nor a globally extended holomorphic structure, selected core, or perturbation spectrum. In particular, the Planck integral and the zero-mode calculation above remain statements about the original patch. The corresponding curved-seed problem has a different answer, as the next calculation shows.

25.4 An exact curved seed can become singular at the end

For a general four-dimensional seed,

$$\widehat{R}(x,y) = \Omega^{-2}(y)R[g](x) + \frac{14(4-s^2)}{\ell^2(1+s^2)}. \tag{25.18}$$

Thus a nonzero slice scalar curvature is amplified as s^2 toward the end. Consider the exact CR solution

$$g = -\mathrm{d}t^2 + (t/t_0)^{2/3}\mathrm{d}\mathbf{x}^2, \qquad \vartheta = \frac{2}{3}\log(t/t_0), \qquad u_j = v_j = 0. \tag{25.19}$$

It has $R[g] = -2/(3t^2)$. At any fixed $t > 0$,

$$\widehat{R} = -\frac{2(1+s^2)}{3t^2} + \frac{14(4-s^2)}{\ell^2(1+s^2)} \longrightarrow -\infty. \tag{25.20}$$

This fixed-time limit establishes unbounded scalar curvature toward normal infinity. By itself it does not prove a singularity at an endpoint reached by an affinely parametrized light ray: the slice time can change along that ray. A separate geodesic test is needed.

For the seed (25.19), keep the slice spatial coordinates and normal angles fixed. An outgoing radial null geodesic obeys

$$t = r + t_*, \qquad k^t = k^r = \frac{E}{\Omega^2}, \qquad \Delta\lambda := \lambda_\infty - \lambda = \frac{\ell}{E}\left[\frac{\pi}{2} - \arctan(r/\ell)\right], \tag{25.21}$$

where $t_* > 0$ and $E > 0$. Here E is conserved along this restricted geodesic; the full time-dependent metric need not have a timelike Killing field. The scalar curvature along the ray is finite: $\widehat{R} \to -44/(3\ell^2)$. Nevertheless the null Ricci contraction is

$$\begin{aligned}\widehat{R}_{MN}k^M k^N &= \frac{3}{2}\left(\frac{\mathrm{d}\vartheta}{\mathrm{d}\lambda}\right)^2 + 6\sum_a \left(\frac{\mathrm{d}N^a}{\mathrm{d}\lambda}\right)^2 \\ &= \frac{2E^2}{3\Omega^4 t^2} + \frac{6E^2}{\ell^2}, \qquad \lim_{r\to\infty}(\Delta\lambda)^2\widehat{R}_{MN}k^M k^N = \frac{2}{3}.\end{aligned} \tag{25.22}$$

The potential term drops out because k is null. Since k is parallel transported along the affinely parametrized geodesic, this divergence is a parallel-propagated curvature singularity at finite affine distance. It obstructs a C^2 Lorentzian extension through this endpoint with a regular continuation of the geodesic. Unlike the fixed-time limit alone, this calculation establishes a genuine extension problem for the lifted scalar seed, even though its Ricci scalar stays finite along the ray.

The field equations remain satisfied at every finite s. Related horizon difficulties occur in warped black-string constructions [13]; the counterexample here follows from the explicit scalar seed and requires no black hole. The flat-seed local extension of Proposition 25.1 does not apply to this rolling-scalar geometry. The divergence in (25.22) prevents a corresponding regular C^2 continuation through its stated endpoint.

A viable global mother theory must control this amplification. Changing the distant normal geometry, allowing the defect profile to respond to slice curvature, or imposing a different global completion changes the problem and may destroy the simple exact ansatz. A finite four-dimensional Planck coefficient is not a substitute for that analysis.

26 Geometric tests on the exact defect

26.1 Its auxiliary geometry and compatible connection

For the flat-seed defect use paired coordinates $z^\mu = x^\mu + \mathrm{i}y^\mu$, $\mu = 0, \ldots, 3$. The label y^0 denotes a positive normal coordinate, not a second time. Let

$$\widehat{g} = \Omega^2 G_0, \quad G_0 = \mathrm{diag}(-1,1,1,1,1,1,1,1), \quad \Omega = \frac{\ell}{\sqrt{\ell^2 + r^2}}, \quad r^2 = \sum_\mu (y^\mu)^2. \tag{26.1}$$

Take $J\partial_{x^\mu} = \partial_{y^\mu}$, $J\partial_{y^\mu} = -\partial_{x^\mu}$, and P tangent to constant-y leaves. Then

$$q = \Omega^2 \mathbf{1}_8, \qquad \eta_{\mathrm{B}} = \Omega^2 \begin{pmatrix} 0 & \mathbf{1}_4 \\ \mathbf{1}_4 & 0 \end{pmatrix}, \qquad \omega = \Omega^2 \omega_0, \quad \omega_0 = -\sum_\mu \mathrm{d}x^\mu \wedge \mathrm{d}y^\mu. \tag{26.2}$$

Both distributions and J are integrable, but

$$\mathrm{d}\omega = 2\mathrm{d}\log\Omega \wedge \omega. \tag{26.3}$$

Away from the core this is nonzero. The positive Hermitian geometry is conformal to a flat Kähler geometry, with exact Lee form $2\mathrm{d}\log\Omega$. Closure of ω would therefore exclude this specific auxiliary geometry unless the variables or background were changed.

The complex-product connection in this coordinate patch is simply $\nabla^{\mathrm{CP}} = \partial$. It has zero ordinary torsion and preserves the constant J, K, while

$$\nabla^{\mathrm{CP}} \widehat{g} = 2\mathrm{d}\log\Omega \otimes \widehat{g}. \tag{26.4}$$

This is explicit nonmetricity. The physical Levi-Civita connection differs by

$$\widehat{\Gamma}^a{}_{bc} = \delta^a_b A_c + \delta^a_c A_b - (G_0)_{bc}(G_0)^{ad} A_d, \qquad A_c = \partial_c \log\Omega. \tag{26.5}$$

The Levi-Civita obstruction proved in the next subsection is not an obstruction to a separate compatible complex connection. In this example that connection is flat even though physical spacetime is curved. An action constructed solely from its curvature would miss physical curvature unless it also accounted for the metric/connection relation.

26.2 Even an unrestricted parallel J is impossible on the flat defect

One might weaken Hermitian compatibility and instead ask only for

$$\widehat{\nabla} J = 0, \qquad J^2 = -\mathbf{1}. \tag{26.6}$$

This still excludes the established flat defect.

Theorem 26.1 (Parallel-complex-structure obstruction)**.** *The metric*

$$\widehat{g} = \frac{\ell^2}{\ell^2 + r^2}\left(\eta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu + \delta_{AB}\mathrm{d}y^A \mathrm{d}y^B\right) \tag{26.7}$$

admits no real Levi-Civita-parallel almost complex structure on a neighborhood of its core. No assumption of $\widehat{g}$-orthogonality of J is needed.

Proof. At the core, choose $G = \mathrm{diag}(-1,1,\ldots,1)$. The first derivatives of $\log\Omega$ vanish; its normal Hessian is $-\delta_{AB}/\ell^2$. The mixed curvature endomorphisms are

$$(R_{\mu A})^a{}_b = \frac{1}{\ell^2}\left(\delta^a_\mu G_{Ab} - \delta^a_A G_{\mu b}\right), \quad \mu = 0,\ldots,3, \quad A = 4,\ldots,7. \tag{26.8}$$

Here a, b run over all eight frame components; the shifted normal labels apply only to this matrix calculation. These sixteen mixed boosts and rotations, together with their commutators, generate all plane generators of $\mathfrak{so}(1,7)$. Any real endomorphism commuting with them is a real scalar multiple of the identity: commuting with spatial plane rotations fixes a scalar spatial block, and the boosts force the time entry to equal it and eliminate the remaining off-diagonal entries.

A parallel J must commute with each curvature endomorphism. It would therefore be $J = a\mathbf{1}$ at the core for real a, incompatible with $J^2 = -\mathbf{1}$. The contradiction is local. The supplied exact matrix calculation independently finds a one-dimensional commutant, spanned by $\mathbf{1}$. □

The standard coordinate J on (26.1) is integrable but is not parallel for the physical metric. Integrability, holomorphicity of a field, and Levi-Civita parallelism are different requirements. A separate complex connection could avoid this particular obstruction, but its relation to the physical connection would then need its own action and equations.

26.3 Why the joint trial action changes this background

The derivative terms in (12.2) cannot be appended while simply reusing the old uplift proof. This can be checked without solving all new equations.

Take the established flat defect with its equatorial X, constant Q, and use the standard constant J exchanging the four x and four y directions. Let P project onto the x directions. All algebraic, alignment, and integrability constraints hold. On this configuration,

$$X^A = \frac{y^A}{\sqrt{\ell^2 + r^2}}, \qquad X^5 = \Omega, \qquad X^6 = 0, \qquad \Lambda = \frac{-6 + 54\Omega^2}{\ell^2}. \tag{26.9}$$

In particular, $\partial_B X^A|_{y=0} = \delta^A_B/\ell$ and the core is regular. The auxiliary metric and warp variable are

$$q = \Omega^2 \delta_8, \qquad A = \log \Omega, \tag{26.10}$$

and a direct connection calculation gives

$$\mathfrak{K}[J] = 16\Omega^{-2}|\nabla_y A|^2, \qquad \mathfrak{K}[P] = 4\Omega^{-2}|\nabla_y A|^2. \tag{26.11}$$

Suppressing the constant external volume and the common prefactor $M_8^6/2$, the added action density is

$$-\kappa_{\mathrm{w}}\, e^{6A}|\nabla_y A|^2, \qquad \kappa_{\mathrm{w}} = 8\kappa_J + 2\kappa_P. \tag{26.12}$$

Now vary the conformal factor A, holding J, P, X, Q fixed. This is an allowed tangent variation of every displayed constraint: orthogonality of P is unchanged by a conformal rescaling, and the remaining constraints do not depend on that factor. Their multiplier terms therefore contribute zero to this variation, even for nonzero multiplier values. The S_* equations vanish on the original defect. The extra Euler residual is

$$\begin{aligned} \mathcal{E}_A^{\mathrm{extra}} &= 2\kappa_{\mathrm{w}} e^{6A}\left(\Delta_y A + 3|\nabla_y A|^2\right) \\ &= 2\kappa_{\mathrm{w}}\Omega^6 \frac{r^2 - 4\ell^2}{(\ell^2 + r^2)^2}. \end{aligned} \tag{26.13}$$

At the core it is $-8\kappa_{\mathrm{w}}/\ell^2$, not zero for positive κ_J, κ_P.

Proposition 26.2 (Failure of the unmodified defect ansatz)**.** *The old flat defect, equipped with the standard constant coordinate J and tangent projector P, is not a solution of* (12.2) *when $\kappa_J, \kappa_P > 0$. Constraint reactions cannot remove the nonzero conformal variation* (26.13)*.*

This does not rule out another J, P configuration, a deformed defect solution, or a different geometric action. It rules out the particular claim that adding natural tensor kinetic terms preserves the old exact background automatically. Taking both couplings zero removes this obstruction but also removes these derivative terms; it is not a derivation of dynamical complex geometry. Cancelling one background residual with signed couplings would still require every other equation and the physical kinetic matrix to be checked.

27 H: the mirror holomorphic parent

The preceding B and D reductions reproduce the same specified potential-free four-dimensional action. H represents its real-analytic solutions in local holomorphic form. It is neither a third stationary point of $\mathcal{V}_\eta$ nor an extra eight-dimensional volume action. The complexification and reflected-conjugation conventions are those of Section 7.

27.1 Fields, measure, and multiplier

The independent variables may be taken as sharp-real holomorphic H, B, Ξ, response fields, and a symmetric multiplier Λ_H^{ab}. Equivalently, H and B describe the independent real slice components of Γ_+. Choose compatible local branches of

$$\mathrm{vol}_H = \sqrt{-\det H}\,\mathrm{d}z^0 \wedge \mathrm{d}z^1 \wedge \mathrm{d}z^2 \wedge \mathrm{d}z^3, \tag{27.1}$$

and the analogous complex metric volumes, with $\mathrm{vol}_H\,|_M = \mathrm{vol}_g$. The holomorphic four-form Lagrangian is

$$\begin{aligned}\mathcal{L}_H = {} & \frac{M_{\mathrm{P}}^2}{4}\left(\mathrm{vol}_{\Gamma_+} R[\Gamma_+] + \mathrm{vol}_{\Gamma_-} R[\Gamma_-]\right) \\ & + \mathrm{vol}_H\, \Lambda_H^{ab}(B_{ab} - \Xi H_{ab}) + \mathcal{L}_{\mathrm{resp}}[H, \Xi, \{q_j\}], \end{aligned} \tag{27.2}$$

$$S_H = \int_M \iota^* \mathcal{L}_H. \tag{27.3}$$

Here $\Lambda_H^\sharp = \Lambda_H$ and $\iota : M \hookrightarrow \mathcal{U}$. The action is real because the pulled-back four-form is real. It integrates a holomorphic $(4,0)$-form over a real four-dimensional contour; it is not an integral over eight real volume coordinates. Suitable boundary terms or the local variational prescription in Appendix A are understood.

Variation of Λ_H imposes (7.4). The multiplier is needed before varying. Substitution of a field restriction into an action need not preserve the equations of an unrelated unrestricted theory [7]. Here the multiplier explicitly defines the restricted theory, and the equivalence to its reduced equations follows by the chain rule. Appendix A gives the relevant local argument.

27.2 A holomorphic response law

For one unnormalized response pair, let Q_1 and Q_2 be sharp-real holomorphic fields and put

$$Z = Q_1 + \mathrm{i}Q_2, \qquad Z^\sharp = Q_1 - \mathrm{i}Q_2, \qquad \rho = (1 + \Xi^2)^{1/2}, \quad \rho|_M > 0. \tag{27.4}$$

The symmetric quadratic form

$$\frac{1}{2}\left[F_+(\mathrm{d}Z)^2 + F_-(\mathrm{d}Z^\sharp)^2\right] + C\rho\,\mathrm{d}Z\,\mathrm{d}Z^\sharp \tag{27.5}$$

has the matrix

$$\begin{pmatrix} C\rho + 1 & -\Xi \\ -\Xi & C\rho - 1 \end{pmatrix} \tag{27.6}$$

in (Q_1, Q_2). Products of differentials here denote symmetric target quadratic forms, to be contracted with H^{ab}. Although the second term resembles a Hermitian expression on the real slice, the reflected operation in (27.5) keeps it holomorphic on $\mathcal{U}$.

Set $U = \sqrt{C+1}\,Q_1$ and $V = \sqrt{C-1}\,Q_2$. With

$$\varphi = f \operatorname{arsinh} \Xi, \qquad \rho = \cosh(\varphi/f), \tag{27.7}$$

the normalized matrix is exactly (9.4), analytically continued. Thus

$$\mathcal{L}_{\text{resp}} = -\frac{1}{2} \operatorname{vol}_H H^{ab} \sum_{j=1}^{n_{\text{P}}} (\partial_a q_j)^{\mathsf{T}} \mathsf{K}_{C_j}(\varphi/f) \partial_b q_j. \tag{27.8}$$

The φ kinetic term is not inserted here: it comes from the curvature term. Positivity is a statement on the physical real slice. A complex quadratic form away from it has no ordering by positive and negative real eigenvalues.

27.3 Exact curvature reduction

Define $Y = H^{ab}\partial_a\Xi\partial_b\Xi$ and $D = \Box_H \Xi$. On the regular constrained branch, the four-dimensional conformal curvature identity gives

$$\operatorname{vol}_{F_\pm H} R[F_\pm H] = \operatorname{vol}_H \left[F_\pm R[H] \mp 3\mathrm{i} D - \frac{3Y}{2F_\pm} \right]. \tag{27.9}$$

Since $F_+F_- = 1 + \Xi^2$, the average is

$$\frac{\operatorname{vol}_{\Gamma_+} R[\Gamma_+] + \operatorname{vol}_{\Gamma_-} R[\Gamma_-]}{2} = \operatorname{vol}_H \left[R[H] - \frac{3Y}{2(1+\Xi^2)} \right]. \tag{27.10}$$

The $\Box_H\Xi$ terms cancel between the two reflected contributions. This is a local density identity on the chosen branches, not merely an equality after dropping that scalar divergence. Standard metric boundary terms remain relevant when defining a finite-domain action.

Using $f^2 = 3M_{\text{P}}^2/2$, the reduced action is

$$S_H^{\text{red}} = \int_M \iota^* \left\{ \operatorname{vol}_H \left[\frac{M_{\text{P}}^2}{2} R[H] - \frac{1}{2} \mathsf{G}_{IJ}(\Psi) H^{ab} \partial_a \Psi^I \partial_b \Psi^J \right] \right\}, \tag{27.11}$$

where $\Psi = (\varphi, U_1, V_1, \ldots)$. Its restriction is precisely (9.1). The construction uses no expansion in Ξ, momentum, or curvature.

Theorem 27.1 (Local analytic parent)**.** *Fix the real structure and regular branches above. The multiplier action (27.3) is locally equivalent to the sharp-real holomorphic extension of (9.1). Every real-analytic solution of the latter admits a unique local holomorphic extension satisfying the reduced equations and the constraint $B = \Xi H$. Conversely, restriction of a sharp-real solution gives a real CR solution.*

Proof. The multiplier chain rule gives equivalence to (27.11). Real-analytic fields extend locally to holomorphic fields. Their real equations are analytic expressions in a finite number of derivatives and inverse nondegenerate metrics; the residuals extend holomorphically and vanish on M. Theorem 7.1 makes the residuals vanish throughout a smaller neighborhood. The same argument gives uniqueness. Reality of the restriction is supplied by the sharp conditions. □

28 Evolution, particles, and the limits of the holomorphic parent

28.1 Field equations and analytic Cauchy data

The extended equations are the holomorphic Einstein–wave-map equations,

$$M_{\mathrm{P}}^2\left(R_{ab}[H]-\frac{1}{2}H_{ab}R[H]\right)=\mathsf{G}_{IJ}\partial_a\Psi^I\partial_b\Psi^J-\frac{1}{2}H_{ab}\mathsf{G}_{IJ}H^{cd}\partial_c\Psi^I\partial_d\Psi^J, \tag{28.1}$$

$$\Box_H\Psi^I+\Gamma^I{}_{JK}(\mathsf{G})H^{ab}\partial_a\Psi^J\partial_b\Psi^K=0. \tag{28.2}$$

For real-analytic constrained Cauchy data on a spacelike three-surface, one can first use a harmonic gauge and the analytic local Cauchy theorem on the noncharacteristic system, then extend the resulting real-analytic solution. Equivalently, the data may be complexified and the analytic equations solved locally. The Cauchy–Kowalevsky theorem concerns analytic, noncharacteristic data [8]; it does not imply that arbitrary smooth fields become analytic.

The gauge constraint propagation equations follow from the contracted Bianchi identity and the wave-map equations. With compatible initial gauge data they retain the Einstein constraints. Reflection maps an analytic solution to a solution with the same real data; local uniqueness preserves the sharp-real conditions. On M, the physical principal symbol of the metric and scalar equations is governed by $g^{\mu\nu}\xi_\mu\xi_\nu$. This is the same characteristic light cone as in the slice theory.

Smooth local evolution and analytic extension have different hypotheses. A smooth solution need not have any holomorphic extension. No claim about global analytic continuation, singularity avoidance, or a global eight-dimensional Cauchy problem follows from Theorem 27.1.

28.2 A real Hamiltonian flow inside complex phase space

On the holomorphic cotangent bundle, write

$$\mathcal{Q}_H=H^{ab}P_aP_b+m^2,\qquad \mathcal{Q}_\pm=F_\pm^{-1}\mathcal{Q}_H. \tag{28.3}$$

The real fixed locus has $z=x$, $P=p\in T_x^*M$. The holomorphic symplectic form and $\mathcal{Q}_H$ respect the lifted real structure. Hamilton's equations for $\mathcal{Q}_H/2$, with real parameter, therefore have a vector field tangent to this real locus. Uniqueness preserves real positions and real momenta, and the restriction is the flow of $\mathcal{Q}_{\mathrm{R}}/2$.

Both complex representatives have the same vanishing set on the real locus. The momenta P_a are independent fiber coordinates on the holomorphic cotangent bundle; extending a single particle trajectory would not determine a covector field on a spacetime neighborhood. Along the real flow,

$$\frac{\mathrm{d}\mathcal{Q}_\pm}{\mathrm{d}\tau}=F_\pm^{-1}\frac{\mathrm{d}\mathcal{Q}_H}{\mathrm{d}\tau}-F_\pm^{-2}\frac{\mathrm{d}F_\pm}{\mathrm{d}\tau}\mathcal{Q}_H=0 \tag{28.4}$$

on the mass shell. One must not replace the chosen real flow by an independent complex evolution parameter without defining a different physical prescription. P2 is preserved by the specified flow, not by every possible use of the same quadratic equation.

28.3 Why the contour does not select itself

A holomorphic top form on a complex four-manifold is closed:

$$\mathrm{d}\mathcal{L}_H=\partial\mathcal{L}_H+\bar\partial\mathcal{L}_H=0. \tag{28.5}$$

The first term would have degree $(5,0)$, and the second vanishes by holomorphicity. Hold the ambient fields, their derivatives, the reflection data, and hence $\mathcal{L}_H$ fixed. For a contour

deformation generated by V, Cartan's formula yields

$$\delta_V \int_M \iota^* \mathcal{L}_H = \int_M \iota^* \left(\iota_V \mathrm{d}\mathcal{L}_H + \mathrm{d}\iota_V \mathcal{L}_H\right) = \int_{\partial M} \iota^* (\iota_V \mathcal{L}_H). \tag{28.6}$$

Compactly supported contour variations give no local embedding equation. Thus this contour action alone does not explain why the physical observers occupy one particular real slice.

Equation (28.6) is not a variation of the real structure σ. Changing σ changes the sharp operation and the Lagrangian itself; that would require a separate variational theory. Parent H supplies σ, the chosen fixed component, and the boundary prescription as geometric data.

28.4 Complex branch boundaries and unselected interactions

For real finite χ, both F and $1+\chi^2$ are nonzero. Their holomorphic continuations need not remain so:

$$F_+ = 0 \text{ at } \Xi = \mathrm{i}, \qquad F_- = 0 \text{ at } \Xi = -\mathrm{i}, \qquad 1 + \Xi^2 = 0 \text{ at both points.} \tag{28.7}$$

In the φ chart, $\cosh(\varphi/f) = 0$ at

$$\varphi/f = \mathrm{i}\pi(k + \tfrac{1}{2}), \qquad k \in \mathbb{Z}. \tag{28.8}$$

These are degeneracy or branch loci in complex field space. Their preimages in spacetime depend on the solution. They do not define a universal width of the allowed imaginary coordinate region or a new measured length scale.

Holomorphicity also does not select the response law. For example,

$$\widetilde{\mathsf{K}}_C(\varphi/f) = \mathsf{K}_C(\varphi/f) + \varepsilon(\varphi/f)^2 \,\mathrm{diag}(1,0), \qquad \varepsilon > 0, \tag{28.9}$$

is holomorphic, sharp-real, and positive on the real slice, with the same vacuum normalization. It changes interactions while preserving the local Einstein–sigma-model structure. An analytic parent therefore does not remove the interaction freedom already present in the slice theory.

For the deformation in (28.9), the matrix and its first derivative at $\varphi = 0$ are unchanged, but its second derivative is not. In the real dimensionful target, the scalar curvature at the identified reference point shifts by $-2\varepsilon/f^2$. This is a concrete interaction freedom; the same light-cone and mass-shell conditions do not determine the target curvature.

28.5 A finite complex-time degeneration

The rolling-scalar solution (9.9) makes the branch limitation explicit. Its analytic continuation on a chosen logarithm branch gives

$$\Xi(z^0) = \sinh\left[\frac{2}{3}\,\mathrm{Log}(z^0/t_0)\right]. \tag{28.10}$$

At the point

$$z^0/t_0 = \exp(3\pi\mathrm{i}/4) \tag{28.11}$$

on the principal logarithm branch, $\Xi = \mathrm{i}$, so $F_+ = 0$ and $\det \Gamma_+ = 0$. The analytically continued clock metric H is nonsingular at that point; it is the complex conformal metric that degenerates. The reflected point gives $\Xi = -\mathrm{i}$ and degeneracy of Γ_-. A regular real solution therefore need not define a nondegenerate mirror geometry throughout its complex continuation. This example supplies a solution-dependent obstruction, not a universal cutoff in imaginary time.

28.6 Potentials and two different kinds of nonuniqueness

Parent H can include a sharp-real analytic potential through $-\operatorname{vol}_H V(\Psi)$. The reduced field equations then acquire

$$T^{\mathrm{kin}}_{ab} \longmapsto T^{\mathrm{kin}}_{ab} - H_{ab}V, \qquad \Box_H \Psi^I + \Gamma^I{}_{JK}(\mathsf{G})\partial\Psi^J\cdot\partial\Psi^K - \mathsf{G}^{IJ}\partial_J V = 0. \tag{28.12}$$

Here T^{kin}_{ab} denotes the kinetic stress tensor in (28.1). The local analytic construction survives on regular branches. The exact torus and defect uplift theorems in Sections 16 and 20, however, are proved for $V = 0$. They must not inherit an arbitrary potential simply because H permits one.

There is also freedom that changes the action away from the constraint surface without changing its constrained classical equations. Let

$$T_{ab} = B_{ab} - \Xi H_{ab}, \qquad \Delta\mathcal{L} = \alpha \operatorname{vol}_H (T_{ab}T^{ab})^2, \tag{28.13}$$

with indices raised by H and a real constant α. Both this density and its first variation vanish at $T = 0$. The mirror multiplier still imposes $T = 0$, and the reduced classical equations are unchanged. This off-constraint ambiguity differs from (28.9), which changes the physical response interaction. A uniqueness principle would have to state which of these notions it is intended to fix.

The classical equivalence to Einstein gravity with this particular sigma target also fixes the present observational scope. With the same real observables and matter couplings, H by itself supplies no demonstrated classical observable difference from that Einstein–sigma model. Quantum equivalence would require compatible integration contours, measures, auxiliary-field determinants, constraint treatment, and regularization. It does not follow from a classical change of variables alone.

29 How H is related to the propagating bulk

29.1 Why a direct warped identification fails

One might try to remove the auxiliary status by defining the holomorphic metric coefficients directly from the warped tangential bulk metric:

$$\Gamma_+{}^{\mathrm{trial}}_{\mu\nu}(x,y) = \Omega^2(y)(1 + \mathrm{i}\chi(x))g_{\mu\nu}(x). \tag{29.1}$$

Here the proposed entries are coefficients of $\mathrm{d}z^\mu \mathrm{d}z^\nu$ in the fixed adapted holomorphic chart. This prescription is not generally holomorphic in $z = x + \mathrm{i}y$. Every holomorphic component must satisfy

$$\partial_{y^{\mu+1}}\Gamma_+ = \mathrm{i}\partial_{x^\mu}\Gamma_+. \tag{29.2}$$

The left side of (29.2) vanishes at the core for (29.1), while its right side generally does not. Already the first normal derivatives obstruct the identification for nonconstant slice data.

Even a constant slice metric and constant χ do not rescue it. Write their constant product as Γ_0. Then

$$\partial^2_{y^A}\Gamma_+{}^{\mathrm{trial}}\big|_{M_N} = -\frac{2}{\ell^2}\Gamma_0, \qquad \partial^2_{x^{A-1}}\Gamma_0 = 0. \tag{29.3}$$

Holomorphicity would require the first quantity to equal minus the second. Thus the real warp profile cannot be identified with a holomorphic conformal factor by this pointwise prescription, even in the simplest vacuum. The obstruction concerns this identification and the supplied J; it is not a theorem excluding every alternative complex structure or variational coupling.

29.2 The valid analytic reconstruction

In adapted coordinates the valid germ is instead

$$\Gamma_+(x+\mathrm{i}y) = \sum_\alpha \frac{(\mathrm{i}y)^\alpha}{\alpha!}\partial_x^\alpha \left[(1+\mathrm{i}\chi(x))g(x)\right], \tag{29.4}$$

where paired indices are understood. It includes the odd imaginary normal derivatives missing from (29.1). It is defined only within the convergence neighborhood of the analytic slice fields. Its normal jets are determined by those fields, not by independent bulk normal data.

This mismatch has an algebraic counterpart: the real part of a nondegenerate complex bilinear four-metric has real signature $(4,4)$, whereas $\widehat{g}$ has signature $(1,7)$, as shown in Section 5. The defect construction therefore improves slice selection without completing a theory generated by one holomorphic metric alone. A future coupling must supply the required analytic jets and retain an acceptable physical evolution problem; the formula (29.1) cannot do so.

30 The common parent and the different meanings of B, D, H

30.1 Exact containment and conditional representation

The pole and equator theorems place the full B and D theories in one fixed action S_*. Their four-dimensional sectors admit a further common statement. Start with a potential-free CR solution (g, Q). If it is real analytic, its clock metric, scalar, and response fields determine the local sharp-real holomorphic fields (H, Ξ, Ψ). The reflected metrics Γ_+ and Γ_- then solve Parent H's constrained equations. Within the same parent, the same (g, Q) has the product lift of Theorem 16.1 and the defect lift of Theorem 20.1, with the respective Planck normalizations.

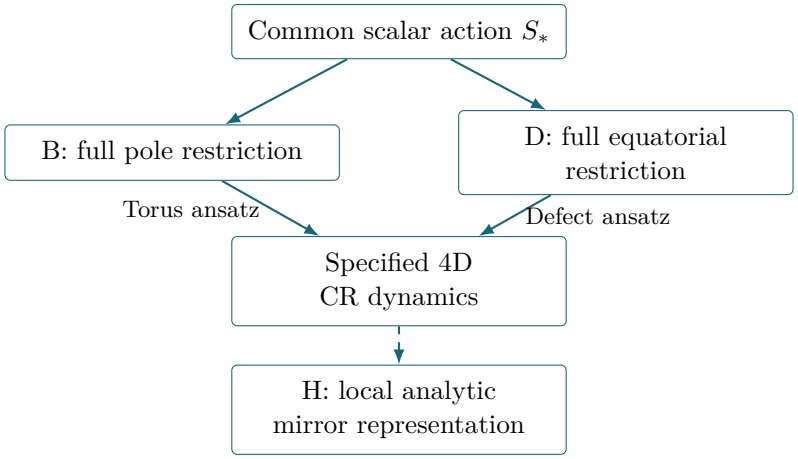


Figure 3: Different arrows assert different results. The upper arrows are complete restrictions of the bulk action. The middle arrows use exact solution ansatzes and their omitted equations. The dashed arrow requires real analyticity and a regular local real structure. The larger geometric trial action is not identified with S_* by this diagram.

Proposition 30.1 (Compatibility through a common restriction)**.** *On their stated regular domains, H, the torus sector of B, and the exact defect sector of D can represent the same analytic four-dimensional solution. Their induced clock metric is g, their complex trace is $(1+\mathrm{i}\sinh\vartheta)g$, and their real free-particle shell is $\mathcal{Q}_{\mathrm{R}} = 0$. B and D are exact restrictions of S_*, but their selected ambient metrics, independent data, and global domains need not coincide. H remains a local representation of the reduced dynamics.*

Proof. Theorem 27.1 provides the analytic germ. The two uplift theorems check all bulk equations separately. The matching multiplier fixes the complex trace and vanishes on shell, so it changes neither bulk solution. On the torus slice and on the defect core, the induced real metric is the prescribed g; in the latter case $\Omega(0) = 1$. The mass-shell statement follows from (1.3). No identity between the two ambient bulk metrics, or between either of them and the realification of Γ_+, is needed. □

In particular, $M_{\mathrm{P}}^2 = M_8^6 \mathcal{V}_4$ for B and $M_{\mathrm{P}}^2 = M_8^6 \pi^2 \ell^4/2$ for D are sector-dependent normalizations in two different geometries of the common local theory. Their right-hand sides are not to be added when one is describing either solution sector. If the two sectors are compared at a common observed M_{P}, their parameters must be chosen accordingly.

The absence of a source from S_{match} is essential here. If S_H were additionally inserted as a core gravitational action with the same trace fields, the holomorphic metric would no longer appear only in the linear matching term. Its equation would no longer set the matching multiplier to zero. Core stress, the embedding equations, and the coefficient of four-dimensional gravity could all change. A thin core of codimension four would also require a distributional or regularized variational prescription. None of the uplift proofs above applies automatically to that different theory.

30.2 Different multipliers, different equations

The original manuscripts used similar symbols for different constraints. In the present notation their roles are:

Multiplier	Equation imposed	Result after variation
Λ_H^{ab}, in H	$B_{ab} = \Xi H_{ab}$	A generally nonzero reaction against the forbidden tensor directions
$\mathcal{M}^{\mu\nu}$, in B or D	$\iota^* \Gamma_+ = (1 + \mathrm{i}\chi) g$	Zero on shell because Γ_+ has no other action in that construction
λ_D, in D	$N^a N^a = 1$	The radial target reaction determined by the sphere equations

Setting $\Lambda_H = 0$ by analogy with $\mathcal{M} = 0$ would change Parent H. Interpreting $\mathcal{M}$ as the material stress holding the defect together would also be incorrect. The actual stress of D comes from the gradients and potential of N, as the component equations show. The common-parent multiplier Λ differs from λ_D by the off-sphere shift (18.13); this shift changes no intrinsic target physics.

30.3 What changes when the torus is replaced by the defect

Property	Flat-torus sector B	Warped defect sector D
Normal geometry	Compact, flat T^4	Noncompact warped $\mathbb{R}^4$
Definition of slice	Chosen fixed reflection component	Regular zero set of $N^1, \ldots, N^4$, with $N^5 > 0$
Gravitational integral	Finite proper normal volume	Finite $\int \Omega^6 \mathrm{d}^4 y$, despite infinite proper normal volume
Normal tensor spectrum	Discrete Fourier masses; nonzero levels have a gap	A threshold zero mode and a gapless continuum
Additional scalar issue	Ten unstabilized internal metric moduli	Four defect scalars mixed with metric fluctuations
Particle behavior	Tangent free flow preserved by reflection	Tangent flow preserved; displaced massive free probes repelled
Holomorphic metric	Auxiliary local germ	Auxiliary local germ on a field-defined analytic core

Neither column proves that generic bulk initial data relax to its displayed sector. A consistent truncation is an invariant family of solutions, not a demonstration of attraction to that family. Nor should a Fourier mass gap in one background be imported into the defect: its normal operator has a different domain and asymptotic geometry.

Part V

Other target sectors and real slices

31 Further invariant target subspaces

31.1 Reflection sectors beyond the original equator

Setting any chosen collection of Cartesian target components to zero gives a great subsphere. These are fixed sets of reflection isometries of the round target, and the potential is even in each component. Their kinetic normal force and potential normal force vanish. Directly, every discarded equation (10.7) is zero because $\partial\mathcal{V}_\eta/\partial X^\alpha$ contains X^α. These are exact consistent restrictions of S_*.

All results in this part use the fixed undeformed S_* unless expressly qualified. They are not automatically solutions of S_{trial}. The original D sector is $X^6 = 0$. A different S^4 is

$$X^5 = 0, \qquad s = X^6, \qquad \mathcal{V}\big|_{X^5=0} = \frac{12}{\ell^2}\left[-1 + (\eta+1)s^2 - \eta s^4\right]. \tag{31.1}$$

This sector contains both B poles and the negative vacuum sphere, and has nontrivial configurations with $X^6 \neq 0$, outside D. It supports a five-dimensional target embedding with four independent target coordinates; it is not a fifth physical spacetime dimension.

The original north-core condition $X^5 > 0$ is unavailable on this equator. A possible alternative would use $X^A = 0$, $X^6 > 0$, provided the four gradients are regular; its realization needs a new solution. Simply exchanging X^5 and X^6 in the old defect while keeping its warp factor fails: at the proposed core the potential is zero, whereas the old warped metric has $\widehat{R}_{\mu\nu} = 4g_{\mu\nu}/\ell^2$. The external Einstein equation is already inconsistent there.

Another exact restriction is

$$X^5 = X^6 = 0, \quad \sum_{A=1}^{4} (X^A)^2 = 1, \qquad \mathcal{V} = -\frac{12}{\ell^2}. \tag{31.2}$$

It is an Einstein–sigma theory with an S^3 target and a negative cosmological term. Constant maps are vacua; varying maps carry three scalar modes. This is a new useful specialization of the parent, but it is contained in D and must not be counted as a wholly separate branch outside D.

Keeping only k of $X^1, \ldots, X^4$, together with X^5, X^6, gives a chain of invariant S^{k+1} targets, $0 \le k \le 4$. In particular the $k = 0$ circle has

$$X^5 = \cos\alpha, \quad X^6 = \sin\alpha, \qquad \mathcal{V} = \frac{12}{\ell^2}\left[\cos^2\alpha + \eta\sin^2\alpha\cos^2\alpha\right]. \tag{31.3}$$

These restrictions reduce scalar dynamics. For $k < 4$, the rank of $\mathrm{d}X^A$ is at most k; the original four-scalar core prescription cannot select a regular codimension-four real slice there. A reduced target is not automatically another four-dimensional observer world.

31.2 Why an arbitrary latitude is not a consistent sector

Write $X = (\cos\alpha\, N, \sin\alpha)$, $N \in S^4$. The normal equation is

$$\widehat{\Box}\alpha + \sin\alpha\cos\alpha\,(\partial N)^2 - \frac{1}{6}\partial_\alpha\mathcal{V} = 0. \tag{31.4}$$

At a generic fixed latitude the middle term depends on unrestricted retained derivatives. It cannot be cancelled identically by a potential depending only on fields. The equator and poles have special status; freezing a generic latitude gives an ansatz that must be checked, not a universal consistent truncation. Particular constant maps at mixed latitudes do nevertheless solve all equations, as the next section shows.

32 All constant-target stationary loci

32.1 Orbit variables and completeness

Set

$$a = \sum_{A=1}^{4} (X^A)^2, \quad b = (X^5)^2, \quad t = (X^6)^2, \qquad a + b + t = 1. \tag{32.1}$$

The potential on the target orbit triangle is

$$v := \frac{\ell^2 \mathcal{V}}{12} = 2b - 1 + (\eta + 1)t - \eta t^2. \tag{32.2}$$

At a stationary point, the coefficients of X^A and X^5 in the multiplier equation differ by $48/\ell^2$. Consequently a and b cannot both be nonzero. On $b = 0$, the remaining interior critical value is $t = (\eta + 1)/(2\eta)$. On $a = 0$, it is $t = (\eta - 1)/(2\eta)$. Including endpoints gives the five rows below, with all sign choices and $O(4)$ orbits included. This proves completeness for this potential and the strict parameter range.

Stationary locus	Target coordinates (a, b, t)	$\ell^2 \mathcal{V}_0$, multiplicity
B poles	$(0, 0, 1)$	0; two points
Negative sphere	$(1, 0, 0)$	-12; one S^3
X^5 poles	$(0, 1, 0)$	12; two points
Mixed spheres	$\left(\frac{\eta-1}{2\eta}, 0, \frac{\eta+1}{2\eta}\right)$	$3(\eta - 1)^2/\eta$; two S^3's
Mixed points	$\left(0, \frac{\eta+1}{2\eta}, \frac{\eta-1}{2\eta}\right)$	$3(\eta + 1)^2/\eta$; four points

The negative sphere is the global minimum. Indeed

$$v + 1 = 2b + t + \eta t(1 - t) \geq 0. \tag{32.3}$$

The mixed isolated points are global maxima, found by maximizing (32.2) first over $b \leq 1 - t$. The B poles are local minima of zero value. A generic zero-potential point is not stationary and cannot be frozen to produce a B-type vacuum.

32.2 The complete order-parameter scalar mass matrices

At a constant target point X_0, let

$$\lambda_0 = X_0 \cdot \partial \mathcal{V}(X_0), \qquad \mathsf{M}^2 = \frac{1}{6} \left(\partial^2 \mathcal{V} - \lambda_0 \mathbf{1}\right)\Big|_{T_{X_0} S^5}. \tag{32.4}$$

The factor $1/6$ is fixed by the kinetic coefficient. The resulting five eigenvalues are:

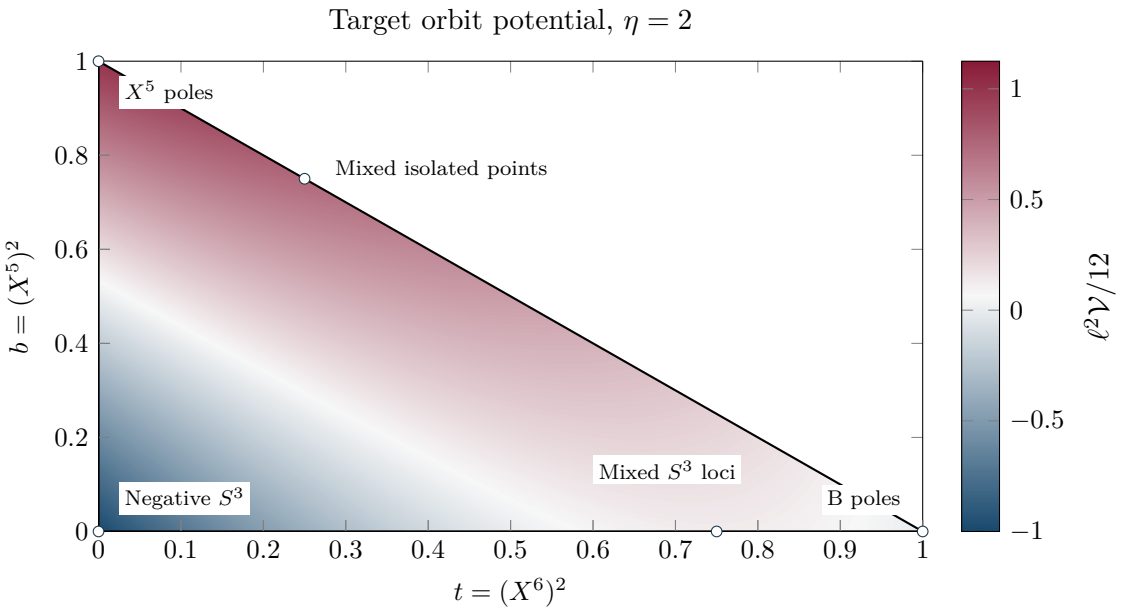


Figure 4: The exact orbit potential at $\eta = 2$. The triangle is $b, t \geq 0$, $b + t \leq 1$. A point represents an $O(4)$ orbit and possible sign choices, rather than necessarily one isolated field configuration. Both mixed stationary classes lie outside the B poles and D equator.

Locus	$\ell^2 m^2$, with multiplicities in parentheses
B poles	$4(\eta - 1)$ (4), $4(\eta + 1)$ (1)
Negative sphere	0 (3), 8 (1), $4(\eta + 1)$ (1)
X^5 poles	-8 (4), $4(\eta - 1)$ (1)
Mixed spheres	0 (3), 8 (1), $-4(\eta^2 - 1)/\eta$ (1)
Mixed points	-8 (4), $-4(\eta^2 - 1)/\eta$ (1)

For example, along $b = 0$, put $t = \sin^2\alpha$. At its interior stationary point, $v''(\alpha) = -2(\eta^2 - 1)/\eta$; multiplying by $12/(6\ell^2)$ gives the negative mixed-latitude mass. The three zero modes run along the vacuum S^3, while the X^5 normal direction has mass $8/\ell^2$.

At a constant map these fluctuations are invariant under linearized spacetime diffeomorphisms and decouple at first order from the metric source, since the tangent potential gradient vanishes. Thus the negative masses give scalar instabilities of the corresponding maximally symmetric de Sitter backgrounds. The nonnegative rows are only scalar-block statements; no nonlinear stability or vacuum-transition calculation is supplied. The CR response fields are constant in these backgrounds and retain their separate massless positive-target fluctuations.

33 Einstein backgrounds and curved four-dimensional products

33.1 A new family of exact bulk backgrounds

For any stationary X_0 and constant Q, equations (10.5)–(10.7) reduce to

$$\mathrm{Ric}(\widehat{g}) = \lambda_8 \widehat{g}, \qquad \lambda_8 = \frac{\mathcal{V}_0}{3}, \qquad \Lambda = \lambda_0. \tag{33.1}$$

Hence every eight-dimensional Einstein metric with this Einstein constant is an exact background of S_*. The maximally symmetric choices are Minkowski for B, anti-de Sitter for the

negative sphere, and de Sitter for each positive row. Their curvature radius obeys $L_8^2 = 21/|\mathcal{V}_0|$. In particular,

$$\begin{aligned} L_8^2 &= \frac{7}{4}\ell^2 && \text{at the negative sphere and the } X^5 \text{ poles,} \\ L_8^2 &= \frac{7\eta\ell^2}{(\eta-1)^2} && \text{at the mixed spheres,} \\ L_8^2 &= \frac{7\eta\ell^2}{(\eta+1)^2} && \text{at the mixed isolated points.} \end{aligned} \tag{33.2}$$

The sign of curvature is supplied by $\mathcal{V}_0$, not by the squared radius. This classification changes when the parent potential is changed. In particular the nonlinear deformations in [3] can move the mixed critical loci while retaining the previously tested B/D restrictions.

33.2 Exact four-dimensional sectors with a cosmological term

A useful further construction is

$$\widehat{g} = g_4(x) \oplus h_4(y), \qquad X = X_0, \qquad Q = Q(x), \tag{33.3}$$

where h_4 is positive definite and

$$\mathrm{Ric}(h_4) = \lambda_8 h_4, \tag{33.4}$$

$$\mathrm{Ric}(g_4)_{\mu\nu} = \mathfrak{h}_{IJ}\partial_\mu Q^I \partial_\nu Q^J + \lambda_8 (g_4)_{\mu\nu}, \tag{33.5}$$

with the usual g_4 wave-map equation for Q. The mixed equations vanish. These equations solve every component of the eight-dimensional system, so this is an exact uplift, not merely a substitution in a reduced action.

For finite internal volume,

$$S_4 = \frac{M_{\mathrm{P}}^2}{2}\int \mathrm{vol}_{g_4}\left[R[g_4] - \mathfrak{h}_{IJ}\partial Q^I \partial Q^J - 2\lambda_8\right], \qquad M_{\mathrm{P}}^2 = M_8^6\,\mathrm{vol}(K_4). \tag{33.6}$$

The internal curvature contributes $R[h_4] = 4\lambda_8$, so the surviving constant is $4\lambda_8 - 2\mathcal{V}_0 = -2\lambda_8$. Internal equations remain essential to this reduction.

With constant Q, local examples are $\mathrm{dS}_4 \times S^4$ for positive λ_8, $\mathrm{AdS}_4 \times H^4$ for negative λ_8, and Ricci-flat products for zero λ_8. At an X^5 pole, both positive-curvature radii satisfy $r_4^2 = 3\ell^2/4$. These are exact classical sectors, but the de Sitter example inherits negative scalar masses, and no full radion or Kaluza–Klein stability analysis is made. Analytic four-dimensional metrics can be complexified locally; equivalence to the potential-free H action would require an appropriate potential completion and is not asserted.

33.3 A global filter on product polarizations

An Einstein product does not automatically admit the proposed global (J, P) structure with its natural product polarization. Here $E = \pi_M^* TM_4$ and $F = \pi_K^* TK_4$, where π_M, π_K are the product projections. The exchange condition requires a bundle isomorphism $E \simeq F$. Restricting to $\{x\} \times K_4$, the first bundle is the trivial bundle $K_4 \times T_x M_4$. Therefore TK_4 must be trivial. Restriction to $M_4 \times \{y\}$ likewise requires TM_4 to be trivial. These are necessary conditions for this global product polarization; they do not by themselves produce an integrable J or solve any additional geometric equations.

For example TS^4 is not trivial: $\chi(S^4) = 2$ obstructs even one nowhere-zero tangent vector. Thus the exact $\mathrm{dS}_4 \times S^4$ background cannot retain that *global product projector* together with

a globally exchanging J. Local choices remain possible, and this argument does not exclude a different nonproduct polarization of the total tangent bundle. It is a restriction on the specified geometric lift, not on the Einstein solution itself.

Corollary 33.1 (Closed internal Einstein spaces must be flat). Assume in addition that K_4 is compact without boundary and that h_4 is a smooth positive-definite Einstein metric. If the global product polarization admits an exchanging J, then h_4 is flat and $\lambda_8 = 0$.

Proof. Triviality of TK_4 supplies a nowhere-zero vector field, so its Euler characteristic $\chi_{\mathrm{E}}(K_4)$ vanishes. With full tensor contractions, the four-dimensional Chern–Gauss–Bonnet formula [21] is

$$32\pi^2\chi_{\mathrm{E}}(K_4) = \int_{K_4} \mathrm{vol}_{h_4} \left(R_{abcd}R^{abcd} - 4R_{ab}R^{ab} + R^2 \right). \tag{33.7}$$

For $R_{ab} = \lambda_8 h_{ab}$, the last two terms cancel: $R_{ab}R^{ab} = 4\lambda_8^2$ and $R = 4\lambda_8$. Thus the integral of the nonnegative Riemann norm is zero. Smoothness and positive definiteness give $R_{abcd} = 0$ everywhere, and hence $\lambda_8 = 0$. □

This excludes every nonzero-λ_8 closed internal Einstein product from this particular global geometric lift, including compact hyperbolic quotients as well as the positive-curvature example. It does not invalidate those metrics as solutions of S_*. The proof does not apply to noncompact finite-volume spaces, manifolds with boundary, warped geometries, or a different polarization; their boundary, topology, and field equations require separate analysis. The symbol χ_{E} here is the Euler characteristic, not the CR conformal scalar χ.

34 Real leaves away from the defect core

The flat defect order parameter is

$$X^{\mu+1} = \frac{y^\mu}{\sqrt{\ell^2 + r^2}}, \qquad X^5 = \Omega, \qquad X^6 = 0. \tag{34.1}$$

Its four gradients have rank four at every finite y. The leaves selected by P are all $y = y_0$, while the order-parameter core is the particular leaf $y_0 = 0$. Every finite leaf is maximally real and has a Lorentzian clock metric

$$g_{y_0} = \Omega(y_0)^2 \eta_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu. \tag{34.2}$$

The pointwise readout (6.2) realizes the same cone prescription on each leaf. Being a possible clock slice is therefore weaker than being the selected defect core.

Their extrinsic geometry distinguishes them. With unit normal $n_\mu = \Omega^{-1}\partial_{y^\mu}$, use the convention $\mathrm{II}^{(\mu)}(U,V) = \widehat{g}(\widehat{\nabla}_U V, n_\mu)$. Directly,

$$\mathrm{II}^{(\mu)}_{ab} = -\Omega^{-1}\partial_{y^\mu} \log\Omega \, (g_{y_0})_{ab} = \frac{y_0^\mu}{\ell\sqrt{\ell^2 + r_0^2}} (g_{y_0})_{ab}. \tag{34.3}$$

All leaves are umbilic; only the core is totally geodesic. For a null tangent, $\mathrm{II}(U,U) = 0$, so an intrinsic null geodesic stays an ambient null geodesic. A free massive geodesic launched tangent to an offset leaf instead has initial normal coordinate acceleration

$$\frac{\mathrm{d}^2 y^\mu}{\mathrm{d}\tau^2} = \frac{y_0^\mu}{\ell^2}, \tag{34.4}$$

when τ is proper time. Confining it to the leaf needs additional forces or couplings. Even the totally geodesic core is not thereby transversely stable. The original probe analysis distinguishes this issue from the cone condition [1].

Normal leaves, when they exist, have positive physical metric $\widehat{g}|_F$. They cannot be interpreted as another Lorentzian observer universe merely by exchanging real and imaginary directions. Likewise a null hypersurface has a degenerate induced metric, so it does not support the same inverse-clock-metric mass shell without a different formalism.

A complex two-dimensional submanifold has four real dimensions, but is not an alternative maximally real observer slice: its tangent is J-invariant. More strongly, a $(2,0)$ metric restricted there has complex rank at most two when represented on the complexified real tangent. Equality of its real null cone with that of a nondegenerate Lorentzian four-metric would instead require a nonzero complex multiple of that four-metric, of rank four. Such complex surfaces can be geometric objects of the parent, but cannot satisfy this same clock-and-cone prescription.

35 Tilted real slices: freedom and a rigidity theorem

35.1 Admissible polarizations of the normalized flat model

Keep G_0 and standard J fixed on $\mathbb{R}^8$. A real four-plane E' is admissible for (3.3) when it is Lorentzian, JE' is its positive orthogonal complement, and P' is the corresponding projector.

Proposition 35.1 (Flat polarization space)**.** *Every such plane is*

$$E' = \mathbb{R}\partial_{x^0} \oplus \mathcal{L}, \tag{35.1}$$

where $\mathcal{L}$ is a Lagrangian three-plane in the Euclidean spatial $\mathbb{C}^3$. The unoriented polarization space is $U(3)/O(3)$. In a graph chart it has the form

$$y^0 = 0, \qquad \mathbf{y} = S\mathbf{x}, \qquad S = S^{\mathsf{T}}, \qquad g_{E'} = -\mathrm{d}(x^0)^2 + \mathrm{d}\mathbf{x}^{\mathsf{T}}(\mathbf{1} + S^2)\mathrm{d}\mathbf{x}. \tag{35.2}$$

Proof. For $u \in E'$, orthogonality gives $G_0(u, Ju) = 2x^0y^0 = 0$. On a linear subspace the product of these two linear functionals can vanish identically only if one functional vanishes identically. The Lorentzian condition excludes $x^0 = 0$, so $y^0 = 0$ throughout E'. The remaining orthogonality condition is spatial symplectic isotropy. Its image in $\mathbb{R}^6$ has dimension at most three; since E' has dimension four it contains the full time axis, and that image is Lagrangian. Conversely every such spatial plane works. Unitary frames act transitively with orthogonal stabilizer. In a graph chart isotropy is precisely symmetry of S. □

With constant P', constant J, constant response fields and a B pole, these flat configurations also solve the displayed J/P trial extension in [3]: all extra derivative terms and all constraints vanish with zero extra multipliers. They do not create a regular field-selected core, because $\mathrm{d}X^A = 0$. Moreover the constant polarizations are related by spatial unitary isometries when all fields are transformed, so they are not automatically inequivalent physical vacua.

35.2 Changing the slice while holding the complex metric fixed

The preceding freedom requires changing the readout Γ with P. If the readout is held fixed, the cone postulate is more restrictive.

Theorem 35.2 (Rigidity for the normalized flat pair)**.** *Fix G_0 and*

$$\Gamma_0 = (1 + \mathrm{i}\chi)\left[-(\mathrm{d}z^0)^2 + \sum_{i=1}^{3}(\mathrm{d}z^i)^2\right], \qquad \chi \in \mathbb{R}. \tag{35.3}$$

If a real Lorentzian four-plane T has the property that every $G_0|_T$-null tangent is Γ_0-null, then T is the original real x-plane.

Proof. Write a nonzero null tangent as $v = (t, \mathbf{x}; u, \mathbf{y})$. The physical null equation is $-t^2 + |\mathbf{x}|^2 + u^2 + |\mathbf{y}|^2 = 0$. On such a vector,

$$\operatorname{Re} \frac{\Gamma_0(v, v)}{1 + \mathrm{i}\chi} = -2|\mathbf{y}|^2. \tag{35.4}$$

Complex nullness forces $\mathbf{y} = 0$; its imaginary part then gives $tu = 0$. A nonzero physical null vector has $t \neq 0$, hence $u = 0$. All null tangents of T lie in the original real plane. Since the null tangents span a Lorentzian four-plane, T equals that plane. □

This theorem is for the specified normalized flat pair, not all possible clock operators A. It shows why "rotate the real slice" is not generally a symmetry while every other physical field is kept fixed. In the defect solution alignment is even more restrictive: its rank-four gradients fix the original tangent distribution. A genuinely new tilted defect must change the order parameter and solve the resulting equations. A coordinate transformation of all fields merely gives another description of the same solution.

36 A kinematic geometry with twisted normal directions

The parent algebra permits geometries beyond the product distributions used in B and D. An explicit example uses standard integrable J and the level-set coordinates

$$\xi^\mu = y^\mu - f^\mu(x), \qquad f^1 = \frac{x^1 x^2}{\ell}, \qquad f^0 = f^2 = f^3 = 0. \tag{36.1}$$

The tangent vectors to $\xi =$ constant include

$$E_1 = \partial_{x^1} + \frac{x^2}{\ell}\partial_{y^1}, \qquad E_2 = \partial_{x^2} + \frac{x^1}{\ell}\partial_{y^1}, \tag{36.2}$$

with $E_0 = \partial_{x^0}$, $E_3 = \partial_{x^3}$. They commute. Define $F_\mu = JE_\mu$. Then

$$[F_1, F_2] = \frac{x^2}{\ell^2}\partial_{x^1}, \tag{36.3}$$

which has a nonzero tangent component when $x^2 \neq 0$ and is not in F. The full frame (E_μ, F_μ) is nonsingular; its nontrivial block determinant is $1 + (x^2/\ell)^2$.

Declare this frame orthogonal with $\widehat{g}(E_\mu, E_\nu) = \eta_{\mu\nu}$ and $\widehat{g}(F_\mu, F_\nu) = \delta_{\mu\nu}$. Choose

$$X^{\mu+1} = \frac{\xi^\mu}{\sqrt{\ell^2 + |\xi|^2}}, \qquad X^5 = \frac{\ell}{\sqrt{\ell^2 + |\xi|^2}}, \qquad X^6 = 0. \tag{36.4}$$

All algebraic constraints, $\mathcal{N}_J = 0$, regular alignment, and a maximally real core hold. But $\mathcal{N}_K \neq 0$. This proves that normal twisting is not excluded by those kinematic requirements.

No Einstein or full trial equation has been solved for this example. It is an admissible geometric configuration, not a newly established exact bulk solution or a derivation of a gauge field. A future action would decide whether such normal twisting propagates, is constrained, or is excluded.

Part VI

Synthesis and the remaining dynamical problem

37 What has been established

Construction	Verified status and limitation
Common scalar action S_*	Exact B/D restrictions at fixed couplings; no dynamical J/P equations
Joint J/P trial action	Explicit variation of all geometric fields; the unmodified defect fails and full constraint closure is open
Born forms plus clock operator	Exact algebraic rewriting of the chosen geometric branch
Second equator, S^3, smaller spheres	Exact target restrictions of S_*; smaller rank may lose the four-dimensional core
Five stationary-locus classes	Complete for $\mathcal{V}_\eta$, $\eta > 1$; scalar masses computed
Einstein and curved products	Exact S_* solutions/uplifts; a closed internal Einstein factor must be flat for the stated global product polarization
Offset defect leaves	Slices of one known solution; massive confinement does not follow
Constant tilted flat polarizations	Flat trial-system solutions, usually symmetry related; no regular X-selected core
Twisted normal distribution	Explicit kinematic example; dynamics unproved
H	Conditional analytic representation, not another target vacuum

The mother geometry is therefore substantially larger than the three original labels, but its enlargement has several levels. Some new sectors follow exactly from the already specified potential. Some are different leaves or polarizations of existing geometries. Others are configurations that the proposed geometric variables permit before field equations are imposed.

The connection results sharpen the next dynamical problem. One can now compare a connection preserving the complex-product structure, the compatible Born connection, and the Levi-Civita connection of the physical clock metric. Their differences, nonmetricity, distribution twisting, and derivatives of A are concrete tensorial quantities. They supply candidate invariants and compatibility equations; their existence does not choose their coefficients or prove their kinetic stability. A unique connection cannot by itself eliminate the nonlinear potential ambiguity already demonstrated in [3].

The trial action in Section 12 must also retain its previous qualification: appending its positive-fiber J/P kinetic terms changes the flat defect equations. None of the new S_* solutions is silently promoted to a solution of that larger action. The next substantive step is to choose a precise geometric action class and test these sectors against its full equations. The aim remains exact fundamental dynamics, with no low-energy expansion assumed here.

38 From a common action to a geometric principle

The integrated result is stronger than the original comparison of three constructions: B and D now belong to a common fixed scalar action, and the admissible geometry has an explicit Born-and-clock description. It is also more constrained than a claim of a finished mother theory. The common action's coefficients are chosen; the joint geometric action changes a tested background; and the same postulates and linear tests admit inequivalent nonlinear potentials.

The next problem is to choose a declared class of geometric actions on $(\hat{g}, J, P, X, Q)$, or equivalently on the Born and clock data, and test its full equations. A connection preserving J, K, a Born connection, and the physical Levi-Civita connection are distinct objects with calculable differences. Their difference tensors, nonmetricity, normal-distribution twisting, and derivatives of A provide specific invariants. None of their coefficients is fixed by naming the geometry. In particular, an action depending only on the curvature of the flat complex-product connection would miss the curvature of the displayed defect unless the metric/connection relation also contributes.

The necessary calculations have an order. First classify independent local invariants and the allowed potentials, including terms invisible on the established sectors. Then derive independent constraints and their rank, paying special attention to the loss of core alignment at constant poles. The reduced kinetic form and principal symbol must be obtained after solving the appropriate lapse, shift, and multiplier constraints. Candidate backgrounds must solve those full equations; the old defect cannot simply be reused after adding tensor kinetic terms. Only then can the B, D, and H reductions be tested in the larger action.

Two further problems remain even for the scalar parent. The even defect-metric scalar and vector perturbations need their coupled fluctuation operators and domains. The regular local flat extension must also be reconciled with global mode normalization and admissible boundary conditions. The compact-product obstruction in Corollary 33.1 must be respected when proposing global internal Einstein geometries. The rolling-scalar example already shows that finite-coordinate uplift does not imply a regular global completion for every slice solution.

Finally, the response target and its C_j require a selection principle or distinguishable observables. Test-particle binding is not a backreacting matter theory; gauge fields, fermions, and physical detector couplings require actions of their own. Classical equivalence on a slice does not determine a quantum measure, state space, or constraint quantization. These are defined open tasks within a program seeking exact fundamental dynamics.

The present synthesis therefore supplies a chain of precise claims: P1/P2 restrict real observations; the parent geometry organizes the fields; S_* contains the full B and D restrictions; their analytic exact four-dimensional sectors admit H; and additional sectors expose both the scope and the limits of that construction. Each link carries its own hypotheses. A future mother principle must explain why this chain, its interactions, and its physical sector are selected.

A Local variation in the analytic category

A.1 Why compactly supported holomorphic variations are not used

A nonzero holomorphic field cannot have compact support in a connected open complex domain. A variational derivation that calls for arbitrary compactly supported holomorphic perturbations would therefore be invalid. The local Euler equations in Parent H can instead be defined by analytic continuation of the real-slice variational problem. There is also a direct analytic test-function argument.

Take a real rectangular coordinate box $K \subset M$ whose closure lies inside the analytic chart. For a scalar component of a field variation, use

$$\delta q(x) = w(x)p(x), \qquad w(x) = \prod_{\mu=0}^{3} \left[(x^\mu - a_\mu)(b_\mu - x^\mu)\right]^3, \tag{A.1}$$

where p is an arbitrary real polynomial. These variations extend holomorphically, vanish to sufficient order at the box boundary for the second-derivative metric action, and can be chosen independently component by component. After integration by parts, stationarity gives

$$\int_K E(x)w(x)p(x)\,\mathrm{d}^4x = 0 \tag{A.2}$$

for each real Euler residual E. Polynomial density on the compact box implies $Ew = 0$; since w is nonzero in the interior, $E = 0$. Complex residuals are tested through independent real and imaginary variations. The resulting analytic equations extend to the germ by Theorem 7.1. No compact support in the complex neighborhood has been assumed.

A.2 Multiplier equivalence

Suppress indices and volume factors, absorbing the latter into a multiplier density. Let $S_0[H, B, \Xi, q]$ denote the part without the constraint and write

$$S = S_0 + \int \lambda \cdot (B - \Xi H). \tag{A.3}$$

Its local equations are

$$\begin{aligned} B - \Xi H &= 0, & E_B + \lambda &= 0, \\ E_H - \Xi\lambda &= 0, & E_\Xi - \lambda \cdot H &= 0, \end{aligned} \tag{A.4}$$

along with the response equations. Eliminating $\lambda = -E_B$ gives

$$E_H + \Xi E_B = 0, \qquad E_\Xi + H \cdot E_B = 0. \tag{A.5}$$

These are exactly the chain-rule equations obtained from $S_0[H, \Xi H, \Xi, q]$, since $\delta B = \Xi\delta H + H\delta\Xi$. This proof handles the local equations even though E_H and E_B contain derivatives. It does not assert that the same restrictions are a consistent truncation of an independently varied complex vacuum action without the multiplier.

For Parent B, write a component of the matching residual as $A_R + \mathrm{i}A_I$ and its multiplier as $\lambda_R + \mathrm{i}\lambda_I$. Then

$$\mathrm{Re}\left[(\lambda_R + \mathrm{i}\lambda_I)(A_R + \mathrm{i}A_I)\right] = \lambda_R A_R - \lambda_I A_I. \tag{A.6}$$

Independent variations of the real and imaginary trace of Γ_+ give both multiplier components zero. The matching equations do not leave an unnoticed localized stress.

B Bulk canonical constraints and counting

The dynamical bulk action (14.2) has the standard Einstein–sigma-model constraint structure. This appendix concerns that action. It does not claim a complete Dirac–Bergmann classification of every holomorphic and slice-multiplier variable in (11.2).

On a seven-dimensional spatial hypersurface, let h_{ij} be the spatial metric, π^{ij} its conjugate momentum, and P_I the scalar momenta. The lapse and seven shifts impose

$$\begin{aligned}\mathcal{H}_\perp = &\frac{2}{M_8^6\sqrt{h}}\left(\pi_{ij}\pi^{ij}-\frac{\pi^2}{6}\right)-\frac{M_8^6\sqrt{h}}{2}\,{}^{(7)}R\\ &+\frac{1}{2M_8^6\sqrt{h}}\mathfrak{h}^{IJ}P_IP_J+\frac{M_8^6\sqrt{h}}{2}\mathfrak{h}_{IJ}h^{ij}\partial_iQ^I\partial_jQ^J, \qquad \text{(B.1)}\\ \mathcal{H}_i = &-2D_j\pi^j{}_i+P_I\partial_iQ^I. \qquad \text{(B.2)}\end{aligned}$$

The denominator 6 in the gravitational trace term is $d-1$ for $d=7$ spatial dimensions. Compactly supported smearings, or a boundary-free spatial domain, give

$$\begin{aligned}\{D[\xi],D[\eta]\} &= D[[\xi,\eta]],\\ \{D[\xi],H[N]\} &= H[\mathcal{L}_\xi N],\\ \{H[N],H[M]\} &= D[h^{ij}(N\partial_jM-M\partial_jN)].\end{aligned} \qquad \text{(B.3)}$$

There are eight first-class secondary constraints. With 28 spatial metric components and n scalars, the physical configuration count is

$$\frac{1}{2}\left[2(28+n)-2\times 8\right]=20+n. \qquad \text{(B.4)}$$

The lapse and shift primary pairs do not change this count. In a local sphere chart, S_* adds five positive target coordinates and the potential contribution $M_8^6\sqrt{h}\,\mathcal{V}_\eta$ to $\mathcal{H}_\perp$. The same diffeomorphism algebra yields $25+n$ modes. This statement does not cover the derivative constraints of S_{trial}. Positivity of $\mathfrak{h}$ makes the scalar kinetic matrix invertible on the real branch. Restriction to reflection-invariant or internally constant configurations changes the retained mode content as described in Section 17; it is not an additional universal reduction of the unrestricted local count.

C How the sphere target and potential were obtained

The exact profile can be derived by an inverse construction. Start with a conformal product $\widehat{g}=e^{2A(y)}(g+\delta)$, keep the CR fields independent of y, and seek additional scalar fields whose normal stress supplies the remaining Ricci components. If $h=e^{-A}>0$, then

$$-6(\partial_A\partial_BA-\partial_AA\partial_BA)=6\frac{\partial_A\partial_Bh}{h}. \qquad \text{(C.1)}$$

A convex h gives a positive semidefinite candidate normal target pullback. A positive-definite Hessian is required if this pullback is to serve as a nondegenerate positive metric in the four normal field coordinates. Convexity alone does not suffice: a positive constant h has zero Hessian. Choose

$$h=\sqrt{1+r^2/\ell^2}. \qquad \text{(C.2)}$$

Its Hessian divided by h is exactly (19.8), the pullback of the unit-sphere metric under (19.3). Its tangential and radial eigenvalues in the flat normal coordinates are

$$\frac{1}{\ell^2+r^2}\quad\text{(multiplicity three)},\qquad \frac{\ell^2}{(\ell^2+r^2)^2}\quad\text{(multiplicity one)}, \qquad \text{(C.3)}$$

both strictly positive at every finite radius. The coefficient 6 is $D-2$ from the eight-dimensional conformal curvature formula. The remaining isotropic Einstein equation fixes the potential along the profile to (19.10).

Since $(N^5)^2 = 1/(1+s^2)$, that profile is generated by the globally bounded sphere potential (18.2). Substituting it into the independent sphere equations verifies that the inverse construction has produced a true solution, rather than only a prescribed stress tensor.

In the chart $Y^A = N^A/N^5 = y^A/\ell$, the target metric is

$$6\mathrm{d}N^a\mathrm{d}N^a = 6\left[\frac{\delta_{AB}}{1+Y^2} - \frac{Y_AY_B}{(1+Y^2)^2}\right]\mathrm{d}Y^A\mathrm{d}Y^B. \tag{C.4}$$

With $|Y| = \tan\alpha$, this becomes $6[\mathrm{d}\alpha^2 + \sin^2\alpha\,\mathrm{d}\Omega_3^2]$. The apparent chart boundary at $N^5 = 0$ is regular on the full sphere target. Target regularity alone does not settle the separate spacetime extension problem; Section 25.3 supplies the additional geometric construction for the flat seed.

The choice of h was made to obtain a simple positive target and a finite normalization. Other choices can generate other models. The calculation determines the coefficients for this chosen profile; it does not derive that profile uniquely from P1 and P2.

D Additional analytic checks

D.1 A radial variation with the lapse retained

A useful independent check keeps the radial lapse until after variation. For a flat slice, use

$$\mathrm{d}\widehat{s}^2 = e^{2a(r)}\eta_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu + B_r(r)^2\mathrm{d}r^2 + R_s(r)^2\mathrm{d}\Omega_3^2, \quad N = (\sin\alpha\,\mathbf{n}, \cos\alpha), \tag{D.1}$$

where $\mathbf{n} \in S^3$, $B_r > 0$, and $R_s > 0$ for $r > 0$. After a radial integration by parts, suppressing the common angular volume and gravitational prefactor, the reduced Lagrangian is

$$\begin{aligned} L_{\mathrm{rad}} = {} & \frac{e^{4a}R_s^3}{B_r}\left[12(a')^2 + 24a'\frac{R_s'}{R_s} + 6\left(\frac{R_s'}{R_s}\right)^2 - 6(\alpha')^2\right] \\ & + e^{4a}B_rR_s(6 - 18\sin^2\alpha) - 2e^{4a}B_rR_s^3\mathcal{U}(\alpha), \end{aligned} \tag{D.2}$$

$$\mathcal{U}(\alpha) = \frac{12\cos(2\alpha)}{\ell^2}. \tag{D.3}$$

The coefficients 6 and 18 in the second line come respectively from the unit three-sphere curvature and the angular sigma gradients. Varying a, B_r, R_s, α first, and only then substituting

$$e^a = B_r = \Omega, \qquad R_s = r\Omega, \qquad \alpha = \arctan(r/\ell), \tag{D.4}$$

gives four vanishing Euler expressions, including the radial constraint. Setting B_r before variation would hide that check. An added constant $1/\ell^2$ in $\mathcal{U}$ makes the lapse residual nonzero, providing a simple negative control. This radial calculation is an additional check of the flat background; the unrestricted component proof in Section 20 supplies the general embedding result. Core regularity is checked in Cartesian coordinates, where the polar-coordinate degeneration is absent.

D.2 Curvature invariants

For the flat seed the full eight-dimensional metric is conformally flat. Thus its Weyl tensor vanishes and

$$\widehat{R}_{MNPQ}\widehat{R}^{MNPQ} = \frac{2}{3}\widehat{R}_{MN}\widehat{R}^{MN} - \frac{1}{21}\widehat{R}^2. \tag{D.5}$$

The eigenvalues of $\widehat{R}^{M}{}_{N}$, grouped into four tangential, three normal-angular, and one normal-radial component, are respectively

$$\frac{4(1-s^2)}{\ell^2(1+s^2)}, \qquad \frac{10+2s^2}{\ell^2(1+s^2)}, \qquad \frac{10-4s^2}{\ell^2(1+s^2)}. \tag{D.6}$$

They give (25.1)–(25.3). These eigenvalues are not a substitute for an analysis of curvature in frames parallel-propagated to the end.

D.3 Canonical count

The physical target has dimension $n+4$. In a local sphere chart its kinetic matrix is positive and nonsingular. Seven spatial dimensions give 28 spatial metric components. The eight diffeomorphism constraints are first class in the Einstein–sigma bulk theory, giving

$$\frac{1}{2}\left[2(28+n+4)-2\times 8\right] = 24+n \tag{D.7}$$

configuration degrees of freedom. The unit-vector multiplier description gives the same target count after its algebraic constraints are eliminated. This is a count for the dynamical bulk action. A full raw Dirac–Bergmann classification of the holomorphic matching variables has not been performed; the matching sector is eliminated by its equations before this count is used.

D.4 A radial boundary check

For a cutoff $r = R$, the radial gravitational boundary density per unit slice volume is proportional to

$$\Omega^6 R^3 \left(7A'(R) + \frac{3}{R}\right). \tag{D.8}$$

It tends to zero as $R \to \infty$, and the analogous small-sphere term tends to zero as $R \to 0$. The finite bulk integral in (20.3) is therefore not hiding a nonzero radial Einstein boundary contribution for this ansatz. Usual four-dimensional boundary prescriptions can be supplied independently.

E Source concordance and verification

E.1 Four sources, one logical order

The supplied B/D/H article contributes the postulates, slice action, holomorphic multiplier proof, exact torus and defect uplifts, canonical analysis, mode domains, probe forces, and global-extension tests. The S^5 note contributes the common fixed action, complete pole/equator restrictions, multiplier shift, and added odd-scalar threshold. The geometric-variation note contributes the J/P variables, algebraic readout, joint trial action, nonuniqueness family, and compatibility obstructions. The parent-geometry note contributes the Born-and-clock description, connection distinctions, full target stationary classification, new Einstein products, slice rigidity, and twisted normal example.

Repeated definitions of the response target and common scalar action have been consolidated. The old claim that B and D merely share slice solutions is replaced by the later exact containment in S_*. The distinct global topology of their displayed backgrounds remains explicit. The mirror tensor multiplier is Λ_H, the readout multiplier is $\mathcal{M}$, and the order-parameter multipliers are Λ, λ_D. The positive metric q is never identified with a physical one-time clock metric or with realification of a complex bilinear metric. The full target

classification applies to the undeformed potential at $\eta > 1$; the odd-mode threshold separately examines all real η.

All four attached TeX sources are preserved unchanged in the package, with recorded hashes and a section concordance. This synthesis is a new document. Historical verification counts and revision notes are not copied into its body as new physical results.

E.2 Reproducible scope

The package contains eight inherited mathematical verification suites: 55 parent checks, 52 defect checks, 36 defect-audit checks, 34 integration checks, 50 integrated-audit checks, 40 common-parent checks, 42 geometric-variation checks, and 72 parent-geometry checks. Their total is 381 named assertions, with overlaps in what they test. They are rerun as inherited suites, not counted as 381 independent predictions or new proofs. A separate synthesis check validates source coverage, labels, citations, ordering, and the identities connecting the merged conventions. The generated reports state the current pass/fail totals.

Revision 1.1 adds a separately written audit suite with 49 exact checks and no imports from the inherited suites. It reconstructs the light-cone linear constraints, checks real Hamiltonian rescalings, tests the Born/clock identities in non-adapted coordinates, derives the response-curvature shift from Christoffel jets, recomputes stationary masses by great-circle variations, contracts all eight J/P connection matrices, and checks the curvature commutant, radial Euler residual, normal operators, probe barrier, and null-ray limits. It also checks the component normalization used in (33.7). These 49 assertions are reported separately; several overlap mathematically with earlier checks.

The audit corrects the comparison between spectral and pointwise mass conditions, separates the P2 shell requirement from its chosen real Hamiltonian realization, states the orientation qualification for a global clock vector, and strengthens the closed internal Einstein obstruction. The four supplied source articles remain unchanged. An audit report records the revisions and their scope.

General proofs in the text remain essential. Exact-point curvature checks do not prove identities on arbitrary backgrounds; matrix examples do not prove all geometric statements; and symbolic checks do not establish nonlinear stability, a complete Hamiltonian classification of the larger trial action, a global complex lift, or a quantum theory. The script results and PDF layout checks are documented separately.

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